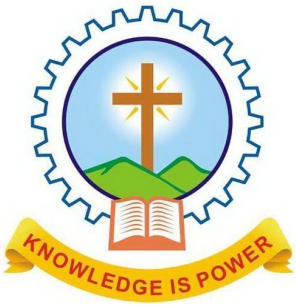
MAR ATHANASIUS COLLEGE OF ENGINEERING

KOTHAMANGALAM

(Government Aided Autonomous Institution)



Department of Computer Applications

Mini Project Report

FAKE REVIEW DETECTION USING MACHINE LEARNING

Done by

ASHISHA S

Reg No: MAC25MCA-2017

Under the guidance of

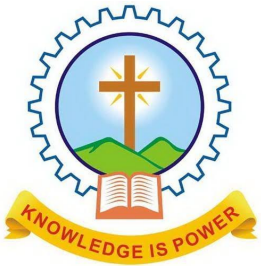
Prof. Biju Skaria

2025–2027

MAR ATHANASIUS COLLEGE OF ENGINEERING

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(Government Aided Autonomous Institution)



CERTIFICATE

Fake Review Detection Using Machine Learning

Certified that this is the bonafide record of project work done by

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during the third semester, in partial fulfilment of requirements for award of the degree,

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Abstract

Online reviews play an important role in influencing consumer decisions and establishing trust in digital platforms. However, the increasing presence of fake and deceptive reviews creates challenges for users in identifying reliable information. This project, Fake Review Detection Using Machine Learning, focuses on developing a machine learning-based system capable of classifying hotel reviews as Fake/Deceptive or Genuine/Truthful based on their textual characteristics.

The system is developed using the Ott et al. Gold Standard Deceptive Opinion Spam dataset, which contains 1,600 labelled hotel reviews. The dataset undergoes exploratory data analysis, text preprocessing, and feature extraction. Term Frequency-Inverse Document Frequency (TF-IDF) is used to represent textual information, while linguistic features and VADER sentiment scores are investigated as additional features for review classification.

Three supervised machine learning algorithms, namely Logistic Regression, Linear Support Vector Machine, and Random Forest, are evaluated using different feature combinations. The models are compared using classification performance measures to identify a suitable configuration for fake review detection.

The experimental results show that Logistic Regression with TF-IDF and VADER sentiment features provides the best overall classification performance, achieving an accuracy of 91.88% on the original test dataset. The selected model is further evaluated using a balanced external dataset of 500 unseen reviews (250 fake and 250 genuine), of which 452 are correctly classified.

A Streamlit-based web application is developed to provide an interactive interface where users can enter hotel reviews and obtain predictions along with class probabilities and explanations of the classification process. The proposed system demonstrates the practical application of machine learning and Natural Language Processing techniques for identifying potentially deceptive online reviews.

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1. Introduction

Online reviews have become an important source of information for consumers when making decisions about products and services. Customers rely on these reviews to evaluate quality, reliability, and overall experiences. However, the growing popularity of online review platforms has also increased the presence of fake or deceptive reviews. Such reviews may be written to promote a business, damage competitors' reputations, or mislead potential customers. This reduces the credibility of online platforms and makes it difficult to distinguish genuine opinions from misleading information.

Machine Learning (ML) and Natural Language Processing (NLP) techniques provide an approach for automatically detecting deceptive reviews. Machine learning algorithms learn patterns from labelled data to classify reviews as fake or genuine, while NLP techniques process and analyse textual information. Feature extraction methods convert review content into numerical representations by considering textual, linguistic, and sentiment-based characteristics.

This project uses the Ott Deceptive Opinion Spam Corpus, a gold-standard dataset containing 1,600 labelled hotel reviews, including 800 deceptive and 800 truthful reviews. The review text undergoes preprocessing techniques such as lowercase conversion, removal of unnecessary characters, tokenization, stop-word removal, and lemmatization. Term Frequency-Inverse Document Frequency (TF-IDF) is used for textual representation. Additional linguistic features, including review length, sentence count, and punctuation-related characteristics, are extracted. VADER sentiment analysis is also applied to obtain positive, negative, neutral, and compound sentiment scores.

Three supervised machine learning algorithms, namely Logistic Regression, Linear Support Vector Machine (SVM), and Random Forest, are considered for classification. Different combinations of textual, linguistic, and sentiment-based features are examined. The models are evaluated using Accuracy, Precision, Recall, F1-score, ROC–AUC, and Confusion Matrix to compare their classification performance.

The main objective of this project is to develop a machine learning-based system capable of identifying fake and genuine hotel reviews. By integrating NLP, feature extraction, sentiment analysis, and supervised machine learning techniques, the system aims to provide a practical approach for detecting deceptive reviews and improving the reliability of online review platforms.

2. Supporting Literature

2.1 Literature Review

Paper I

A. Khan, M. Adnan, and S. Anwar (2025). Empowering Trust in Digital Platforms: Sentiment Analysis and Emotion AI for Fake Reviews Detection. IEEE Transactions on Computational Social Systems, Early Access. doi:10.1109/TCSS.2025.3638615.

Khan et al. (2025) investigated the use of sentiment analysis and linguistic characteristics for detecting fake online reviews. The study highlights that fake and genuine reviews can exhibit different linguistic and emotional patterns. Therefore, sentiment information can be used together with textual and linguistic features to provide additional information for fake review classification.

The study used three publicly available review datasets from different domains, including Android application reviews, hotel reviews, and product reviews. The datasets contain reviews from different application areas, allowing the proposed approach to be evaluated across multiple domains. The study investigated linguistic characteristics such as review length, punctuation, misspellings, adjective usage, and textual similarity along with sentiment-related information.

The methodology includes data preprocessing, linguistic feature extraction, sentiment analysis, machine learning classification, feature importance analysis, and performance evaluation. TF-IDF was used to represent textual information, while sentiment and emotion-related characteristics were extracted to provide additional information about the reviews. Feature analysis was also performed to identify important characteristics contributing to classification.

The study used Random Forest and Decision Tree classifiers for fake review detection. Information gain and approximation-based feature analysis were used for feature selection and analysis, while SHAP was used to interpret the contribution of individual features to the model predictions.

The results demonstrated that integrating linguistic features with sentiment analysis can improve fake review detection performance. The study reported high classification performance for the evaluated datasets, with Random Forest and Decision Tree achieving accuracy of up to approximately 98% with high F1-scores. The findings indicate that sentiment and linguistic characteristics can complement conventional textual information and provide useful signals for identifying deceptive reviews.

The study also highlights the importance of understanding the contribution of individual features to fake review classification. However, the proposed project uses a smaller and more practical feature set that can be directly extracted from the available review text. The proposed

system combines TF-IDF features with linguistic and VADER sentiment features using the Ott Deceptive Opinion Spam Corpus.

Title	Empowering Trust in Digital Platforms: Sentiment Analysis and Emotion AI for Fake Reviews Detection
Authors	A. Khan, M. Adnan, and S. Anwar
Year	2025
Area of Work	Fake review detection using linguistic characteristics, sentiment analysis, feature analysis, and machine learning.
Dataset Used	Three publicly available datasets consisting of Android application reviews, hotel reviews, and product reviews.
Methodology	Data preprocessing, TF-IDF feature extraction, linguistic feature extraction, sentiment and emotion analysis, feature analysis, machine learning classification, SHAP-based interpretation, and performance evaluation.
Algorithms Used	Random Forest and Decision Tree.
Key Findings	The study demonstrated that combining linguistic features with sentiment information can improve fake review detection. The evaluated models achieved high classification performance, with accuracy of up to approximately 98% and high F1-scores.

Table 2.1: Summary of Paper I

Paper II

M. Abdulqader, A. Namoun, and Y. Alsaawy (2022). Fake Online Reviews: A Unified Detection Model Using Deception Theories. IEEE Access, Vol. 10, pp. 128622–128655. doi:10.1109/ACCESS.2022.3227631.

Abdulqader et al. (2022) proposed a unified framework for detecting fake online reviews by combining deception theories with machine learning techniques. The study focuses on identifying characteristics that can distinguish deceptive reviews from truthful reviews. Instead of relying only on conventional textual features, the authors used deception theories to identify both verbal and non-verbal characteristics related to fake reviews.

The proposed methodology consists of several stages, including the selection of deception theories, identification of measurable deception constructs, feature extraction, machine learning-based classification, and experimental evaluation. The authors identified nine important deception constructs, including specificity, quantity, non-immediacy, affect, uncertainty, informality, consistency, source credibility, and deviation in behaviour. These constructs were represented using verbal and non-verbal features obtained from review content and reviewer-related information.

For experimental evaluation, the study used three Yelp benchmark datasets, namely YelpChi, YelpNYC, and YelpZip. The review data were processed and relevant features were extracted before applying machine learning algorithms. Logistic Regression, Naive Bayes, Decision Tree, and Random Forest were evaluated for fake review classification.

The results showed that deception-related constructs were useful for identifying fake reviews. The study found that non-verbal features were important for distinguishing fake reviews and that combining verbal and non-verbal features improved detection performance. The theory-based approach also provided useful interpretability with relatively low computational complexity.

An important limitation of this approach is its dependence on reviewer behavioural and metadata information, which may not always be available in practical review detection systems. This motivates the proposed project to focus mainly on information that can be obtained directly from review content.

Based on these findings, the proposed system focuses on review text and uses textual, linguistic, and sentiment-based features. The proposed project uses the Ott Deceptive Opinion Spam Corpus rather than the Yelp benchmark datasets used in this study.

Title	Fake Online Reviews: A Unified Detection Model Using Deception Theories
Authors	M. Abdulqader, A. Namoun, and Y. Alsaawy
Year	2022
Area of Work	Theory-driven fake review detection using verbal and non-verbal deception features combined with machine learning.
Dataset Used	YelpChi, YelpNYC, and YelpZip benchmark datasets.
Methodology	Selection of deception theories, deception construct identification, verbal and non-verbal feature extraction, machine learning classification, and experimental evaluation.
Algorithms Used	Logistic Regression, Naive Bayes, Decision Tree, and Random Forest.
Key Findings	Deception-related features were useful for identifying fake reviews. Non-verbal features were important, and combining verbal and non-verbal features improved fake review detection performance.

Table 2.2: Summary of Paper II

Paper III

S. Janokar, K. Joshi, K. More, P. Pawar, R. Raikwar, and T. Singh (2024). Fake Review Detection System: A Machine Learning Approach with Linguistic and Semantic Analysis. Proceedings of the 2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAIQSA). doi:10.1109/ICAIQSA64000.2024.10882276.

Janokar et al. (2024) proposed a machine learning-based Fake Review Detection System that incorporates linguistic and semantic analysis for identifying fake reviews. The study emphasizes that fake review detection should consider not only the textual characteristics of reviews but also non-linguistic behavioural characteristics of reviewers.

The study used review data containing authentic and counterfeit reviews obtained from an online review platform. Yelp review data were considered in the study because Yelp's review filtering system provides categorized review information that can be used for fake review detection research. The data were preprocessed through text normalization and removal of irrelevant characters. Techniques were also applied to handle class imbalance and provide a more balanced representation of genuine and fake reviews.

Natural Language Processing techniques were used to extract linguistic and semantic features from the review text. TF-IDF and word embeddings were used to represent textual and semantic information. The study also developed custom linguistic and semantic analysis components to capture subtle characteristics that may indicate deceptive reviews.

Several classical machine learning algorithms were examined, including Random Forest, Support Vector Machine, Decision Tree, and Logistic Regression. Hyperparameters were adjusted to improve classification performance, and cross-validation was used to evaluate model robustness and reduce the possibility of overfitting.

The study evaluated the detection system using Accuracy, Precision, Recall, F1-score, AUC, and Confusion Matrix. The reported model achieved approximately 99.97% accuracy, 99.94% F1-score, 99.97% recall, and 99.97% AUC. The confusion matrix also showed a high number of correct predictions with comparatively fewer incorrect predictions.

The study further examined the relationship between user behavioural features, textual features, and model confidence. The authors reported that non-linguistic behavioural characteristics were important for fake review detection and that combining behavioural and textual information could improve the effectiveness of the detection system.

The findings of this study provide motivation for the proposed project to use linguistic and semantic information for fake review detection. However, the proposed project focuses specifically on review text and does not depend on reviewer behavioural information. TF-IDF features are combined with six linguistic features and four VADER sentiment features to represent the reviews. Logistic Regression, Linear SVM, and Random Forest will be used for classification.

The models will be evaluated using Accuracy, Precision, Recall, F1-score, ROC–AUC, and Confusion Matrix.

Title	Fake Review Detection System: A Machine Learning Approach with Linguistic and Semantic Analysis
Authors	S. Janokar, K. Joshi, K. More, P. Pawar, R. Raikwar, and T. Singh
Year	2024
Area of Work	Machine learning-based fake review detection using linguistic, semantic, and non-linguistic behavioural features.
Dataset Used	Online review data containing authentic and counterfeit reviews, with Yelp review data considered for fake review detection.
Methodology	Data preprocessing, handling of imbalanced data, linguistic and semantic feature extraction, TF-IDF, word embeddings, machine learning classification, hyperparameter adjustment, cross-validation, and performance evaluation.
Algorithms Used	Random Forest, Support Vector Machine, Decision Tree, and Logistic Regression.
Key Findings	The study reported that combining textual and non-linguistic behavioural characteristics can improve fake review detection. The reported model achieved approximately 99.97% accuracy, 99.94% F1-score, 99.97% recall, and 99.97% AUC.

Table 2.3: Summary of Paper III

2.2 Literature Review Summary

The reviewed literature demonstrates that machine learning and Natural Language Processing techniques can be effectively used for detecting deceptive online reviews. The studies explore different combinations of textual, linguistic, sentiment-based, semantic, and behavioural features to distinguish fake reviews from genuine reviews. The literature also shows that combining multiple feature types can provide useful information for classification beyond textual information alone.

Table 2.4 presents a summary of the reviewed research papers, highlighting their methodologies and major contributions.

Paper Title	**Summary**
Paper I Empowering Trust in Digital Platforms: Sentiment Analysis and Emotion AI for Fake Reviews Detection	The study investigates fake review detection using Android application, hotel, and product review datasets. It examines linguistic characteristics such as review length, punctuation, misspellings, adjective usage, and textual similarity, along with sentiment and emotion-related features. Random Forest and Decision Tree algorithms are used for classification. The study highlights the usefulness of combining linguistic and sentiment information and employs SHAP to interpret the contribution of individual features to model predictions.
Paper II Fake Online Reviews: A Unified Detection Model Using Deception Theories	The research introduces a theory-driven approach to fake review detection using the YelpChi, YelpNYC, and YelpZip datasets. It combines verbal deception features derived from review text with non-verbal features associated with reviewer behaviour. Logistic Regression, Naive Bayes, Decision Tree, and Random Forest algorithms are evaluated. The study demonstrates the importance of combining verbal and non-verbal deception indicators for improving fake review detection.

Paper Title	Summary
Paper III Fake Review Detection System: A Machine Learning Approach with Linguistic and Semantic Analysis	The study proposes a machine learning-based fake review detection system using Yelp review data. It incorporates linguistic and semantic information through TF-IDF, word embeddings, and non-linguistic behavioural characteristics. Random Forest, Support Vector Machine, Decision Tree, and Logistic Regression algorithms are used for classification. The research demonstrates the effectiveness of combining linguistic, semantic, and behavioural features using classical machine learning techniques.

Table 2.4: Summary of Reviewed Literature

2.3 Findings and Proposals

The literature review highlights several important findings that guide the development of the proposed Fake Review Detection System. The major findings are summarized below:

1. **Feature Combination:** The reviewed studies indicate that combining different types of features can improve fake review detection. Textual, linguistic, sentiment-based, semantic, and behavioural characteristics capture different aspects of review content. Therefore, the proposed system will examine different combinations of TF-IDF, linguistic, and sentiment-based features to determine their effectiveness in classification.

2. **Linguistic Features:** Previous studies suggest that linguistic characteristics, such as review length, word usage, and punctuation, can provide useful information for identifying deceptive reviews. The proposed system will investigate six linguistic features: Review Length, Sentence Count, Average Word Length, Exclamation Count, Question Count, and Capital Count.

3. **Sentiment Analysis:** Sentiment-related characteristics can provide additional information about the emotional patterns expressed in reviews. Therefore, VADER sentiment analysis will be used to extract four sentiment features: Positive, Negative, Neutral, and Compound scores.

4. **Classical Machine Learning:** The reviewed studies demonstrate the effectiveness of classical machine learning algorithms for fake review classification. Therefore, Logistic Regression, Linear Support Vector Machine (SVM), and Random Forest will be considered for the proposed system.

5. **Practical Feature Selection:** Some existing approaches depend on reviewer behavioural information or metadata, which may not always be available. Therefore, the proposed system will focus on features that can be extracted directly from review text.

Based on these findings, this project proposes a machine learning-based Fake Review Detection System using the Ott Deceptive Opinion Spam Corpus. The review text will undergo NLP pre-processing, including HTML tag removal, lowercase conversion, removal of URLs, numbers, punctuation and special characters, tokenization, stop-word removal, and lemmatization.

TF-IDF with unigram and bigram features will be used for textual representation. Six linguistic features and four VADER sentiment features will also be extracted to analyse writing characteristics and sentiment information. Four feature configurations will be considered: TF-IDF alone, TF-IDF with linguistic features, TF-IDF with VADER sentiment features, and TF-IDF with both linguistic and VADER features.

These feature configurations will be evaluated using Logistic Regression, Linear SVM, and Random Forest classifiers. Model performance will be compared using Accuracy, Precision, Recall, F1-score, ROC–AUC, and Confusion Matrix to identify a suitable classification approach for the proposed system.

3. System Analysis

System analysis defines the framework of the proposed Fake Review Detection System by studying the dataset, identifying its important characteristics, and determining the preprocessing, feature extraction, and machine learning techniques required for fake review classification. This chapter presents the analysis of the Ott gold-standard dataset and describes the proposed preprocessing and feature extraction approach.

3.1 Analysis of Dataset

3.1.1 About the Dataset

The proposed Fake Review Detection System uses the Ott Deceptive Opinion Spam Corpus, a gold-standard dataset developed for the study of deceptive online reviews. The dataset contains labelled hotel reviews and is used for binary classification of deceptive and truthful reviews. The review text will be the primary source of information for Natural Language Processing (NLP), while the rating and class label will be considered during data analysis and feature extraction.

The dataset contains 1600 hotel reviews and three relevant attributes: `Ori_Review`, `Rating`, and `Label`. The `Label` attribute is the target variable, representing the two classes of reviews. The dataset consists of 800 deceptive reviews and 800 truthful reviews, resulting in an equal distribution between the two classes.

The dataset contains textual and numerical attributes. The review text stored in `Ori_Review` will be processed using NLP techniques and represented using TF-IDF. Additional features such as review length, sentence count, average word length, and sentiment scores will also be considered to capture different characteristics of the reviews. VADER sentiment analysis will be used to extract positive, negative, neutral, and compound sentiment features.

The dataset contains an equal number of deceptive and truthful reviews, making it suitable for binary classification without requiring additional class balancing techniques. The extracted features will be combined and used to train Logistic Regression, Linear Support Vector Machine (SVM), and Random Forest classification models.

Property	Description
Dataset Name	Ott Deceptive Opinion Spam Corpus
Total Reviews	1600
Number of Attributes	3
Problem Type	Binary Classification
Target Variable	`Label`
Target Classes	Deceptive (Fake), Truthful (Genuine)
Data Types	Numerical and Textual
Primary Feature	`Ori_Review` (Hotel Review Text)

Table 3.1: Characteristics of the Dataset

Attribute	Description
`Ori_Review`	Hotel review text used as the primary input for NLP and TF-IDF feature extraction.
`Rating`	Numerical rating associated with the hotel review.
`Label`	Target class indicating whether the review is deceptive or truthful.

Table 3.2: Description of Dataset Attributes

The dataset is particularly suitable for the proposed system because it provides manually labelled deceptive and truthful hotel reviews, allowing supervised machine learning algorithms to learn patterns associated with deceptive review writing. The balanced distribution of the two classes also allows the performance of the classification models to be compared using multiple evaluation measures without the results being dominated by one class.

3.1.2 Explore the Dataset

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, and characteristics of the Ott gold-standard dataset before model development. Basic operations such as `shape`, `info()`, `dtypes`, `describe()`, missing value analysis, duplicate detection, and class distribution analysis were used to examine the dataset.

The dataset contains 1600 hotel reviews with four attributes. The `Ori_Review` attribute contains the original review text and serves as the primary source of information for classification. The `Rating` attribute provides the numerical rating associated with each review, while the `Label` attribute represents the target class. The `Unnamed: 0` attribute is an index column and is not used as a feature during model training.

The class distribution was examined to determine whether the dataset required any balancing technique. The dataset contains 800 deceptive reviews and 800 truthful reviews, resulting in an

equal distribution between the two classes. Therefore, no additional class balancing technique was required. The results of the exploratory analysis were used to guide the subsequent preprocessing and feature extraction stages.

```
Dataset shape: (1600, 4)
```

Figure 3.1: Shape of the Dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1600 entries, 0 to 1599
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Unnamed: 0  1600 non-null   int64
 1   Label       1600 non-null   int64
 2   Rating      1600 non-null   int64
 3   Ori_Review  1600 non-null   object
dtypes: int64(3), object(1)
```

Figure 3.2: Dataset Information

```
Data Types of Dataset Attributes:

Unnamed: 0     int64
Label          int64
Rating         int64
Ori_Review    object
dtype: object
```

Figure 3.3: Data Types of Dataset Attributes

```
Missing Values:
Unnamed: 0    0
Label         0
Rating        0
Ori_Review    0
dtype: int64
```

Figure 3.4: Missing Value Analysis

```
Number of duplicate rows: 0
No duplicate rows are present in the dataset.
```

Figure 3.5: Duplicate Record Analysis

	Rating	Label
count	1600.000000	1600.000000
mean	3.000000	0.500000
std	2.000625	0.500156
min	1.000000	0.000000
25%	1.000000	0.000000
50%	3.000000	0.500000
75%	5.000000	1.000000
max	5.000000	1.000000

Figure 3.6: Statistical Summary of Numerical Attributes

```
Review Class Distribution:
Label
0    800
1    800
Name: count, dtype: int64

Deceptive Reviews: 800
Truthful Reviews: 800
```

Figure 3.7: Class Count Summary of Review Labels

3.2 Data Preprocessing

Data preprocessing is an important stage in the development of a machine learning model because the quality of the input data can affect the performance of the classification model. Online reviews may contain HTML tags, different text formats, punctuation, common words, and other textual variations. Therefore, the review text needs to be cleaned and converted into a suitable format before feature extraction and model training.

In the proposed Fake Review Detection System, the preprocessing stage mainly consists of data cleaning, feature engineering, and text feature representation. The review text is first cleaned using Natural Language Processing (NLP) techniques. After cleaning, selected linguistic and sentiment-based features are extracted from the reviews. The textual information is then represented numerically using Term Frequency-Inverse Document Frequency (TF-IDF). These features are combined and selected as inputs for the machine learning models.

3.2.1 Data Cleaning

Data cleaning is the first stage of preprocessing and is performed to remove unnecessary variations from the review text. The objective is to obtain a consistent representation of the reviews while retaining the meaningful textual information required for classification.

In the proposed system, the review text stored in the `Ori_Review` attribute is processed through

a sequence of NLP operations. The implemented cleaning steps include HTML tag removal, lowercase conversion, removal of punctuation and special characters, tokenization, stop-word removal, and lemmatization. The resulting cleaned text is then used for subsequent feature extraction and TF-IDF vectorization.

HTML Tag Removal

HTML tags present in the review text are removed to eliminate unnecessary markup and retain only the actual textual content of the review. This step helps ensure that HTML elements do not become part of the textual features used during classification.

Lowercase Conversion

All review text is converted into lowercase to maintain consistency in the text data. This prevents words with different letter cases, such as “Excellent” and “excellent”, from being treated as different terms during feature extraction.

Removal of Punctuation and Special Characters

Punctuation marks and special characters are removed from the review text to reduce unnecessary textual variations. This produces a simpler representation of the review text for subsequent NLP operations.

Tokenization

Tokenization divides the cleaned review text into individual words or tokens. This allows the text to be processed at the word level during subsequent preprocessing operations.

Stop-word Removal

Commonly occurring words that provide limited useful information for classification are removed from the tokenized reviews. Examples include words such as “the”, “is”, “and”, and “of”. Removing these words helps reduce unnecessary terms before feature extraction.

Lemmatization

Lemmatization converts words into their base or dictionary form. It helps reduce different forms of the same word and provides a more consistent representation of the review text. The lemmatized tokens are then combined to obtain the final cleaned review text.

3.2.2 Analysis of Feature Variables

After text preprocessing, feature extraction was performed to convert the processed reviews into numerical representations suitable for machine learning. The feature set consists of TF-IDF features, linguistic features, and sentiment-based features extracted using VADER.

TF-IDF Features

Term Frequency-Inverse Document Frequency (TF-IDF) was used to represent the textual content of the reviews numerically. Both unigram and bigram features were extracted to capture individual words as well as meaningful two-word combinations. The TF-IDF representation was configured with a maximum of 10,000 features, a minimum document frequency of 2, and a maximum document frequency of 0.95. Sublinear term frequency scaling was also applied to reduce the influence of very frequent terms.

Linguistic Features

Additional linguistic features were extracted from the review text to capture structural and stylistic characteristics of the reviews. The following six linguistic features were used:

- **Review Length:** Represents the total length of a review in terms of words.
- **Sentence Count:** Represents the number of sentences contained in the review.
- **Average Word Length:** Represents the average number of characters per word in the review.
- **Exclamation Count:** Represents the number of exclamation marks present in the review.
- **Question Count:** Represents the number of question marks present in the review.
- **Capital Count:** Represents the number of uppercase characters present in the review.

These features help capture differences in writing style and structural patterns between fake and genuine reviews.

VADER Sentiment Features

VADER (Valence Aware Dictionary and sEntiment Reasoner) was used to extract sentiment-related characteristics from the reviews. Four sentiment features were obtained:

- **Positive:** Represents the proportion of text associated with positive sentiment.

- **Negative:** Represents the proportion of text associated with negative sentiment.
- **Neutral:** Represents the proportion of text associated with neutral sentiment.
- **Compound:** Represents the overall normalized sentiment score of the review, ranging from -1 to +1.

The TF-IDF, linguistic, and VADER sentiment features are combined to form the final feature representation used for the proposed classification models.

3.2.3 Analysis of Class Variables

The target variable of the Ott Deceptive Opinion Spam Corpus is the `Label` attribute, which represents the class of each review. The problem is formulated as a binary classification task in which each review belongs to either the fake or genuine class.

The class labels used in the dataset are defined as follows:

- Fake / Deceptive Review (Label = 1)
- Genuine / Truthful Review (Label = 0)

The dataset contains an equal number of fake and genuine reviews. There are 800 reviews labelled as fake or deceptive and 800 reviews labelled as genuine or truthful, giving a total of 1600 reviews. Therefore, each class represents 50% of the dataset.

Class	Number of Reviews	Percentage
Fake / Deceptive Reviews	800	50%
Genuine / Truthful Reviews	800	50%
Total Reviews	1600	100%

Table 3.3: Distribution of Class Variables

The equal distribution of the two classes indicates that the dataset is balanced with respect to the target variable. Therefore, additional class balancing techniques such as oversampling or undersampling are not required for the proposed system.

3.3 Data Visualization

Data visualization was performed to analyse the distributional characteristics of the Ott gold-standard dataset as well as the properties of the extracted linguistic and sentiment feature space. Visualizing data distributions prior to and following feature extraction provides critical insights into class balance, text length variations, sentiment orientations, and inter-feature correlations.

Dataset Distribution Visualizations

The class distribution and review length variations across the dataset were analysed to examine the overall structure and composition of the raw review data.

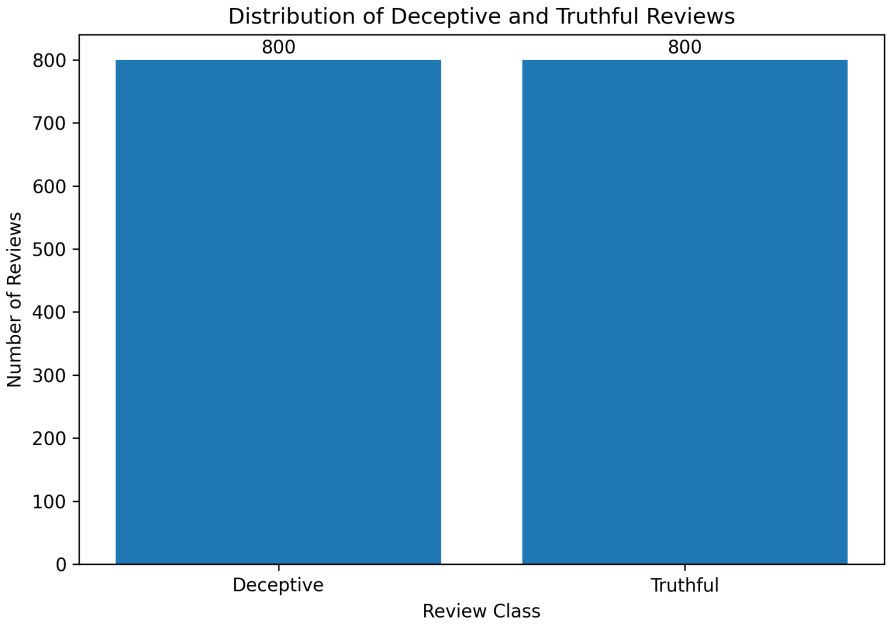


Figure 3.8: Distribution of Deceptive and Truthful Reviews

Figure 3.8 illustrates the distribution of deceptive and truthful reviews in the dataset. The dataset contains 800 reviews in each class, confirming a balanced distribution suitable for unbiased model training.

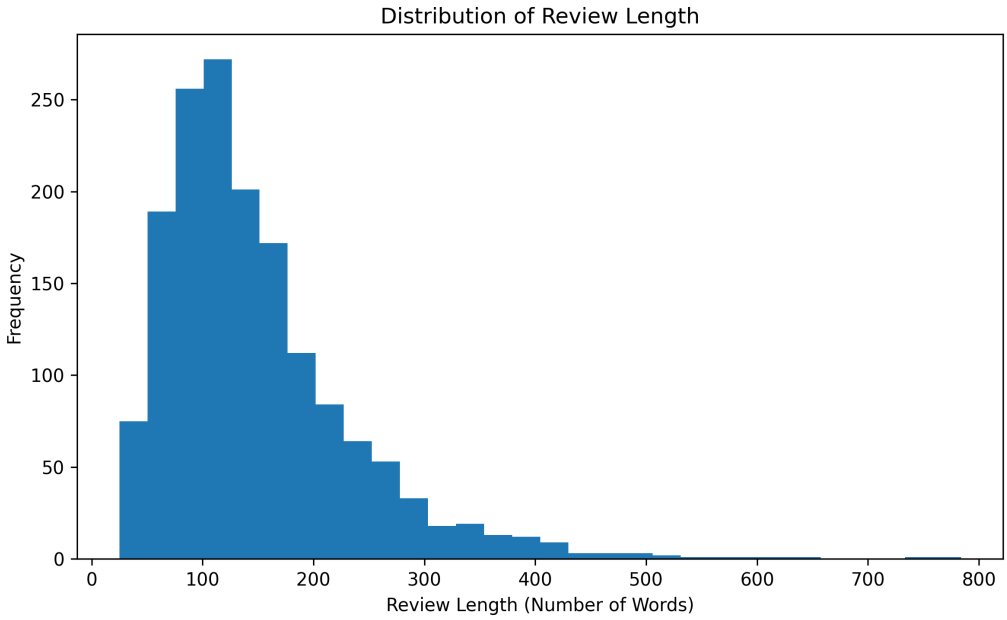


Figure 3.9: Distribution of Review Length

Figure 3.9 displays the variation in review lengths across the corpus. The distribution shows the diversity in word counts between reviews, supporting the inclusion of review length and sentence count as structural linguistic features.

Engineered Feature Visualizations

Following data cleaning and feature engineering, visual analyses were conducted on the VADER sentiment scores and engineered linguistic attributes to assess their discriminatory characteristics between fake and genuine reviews.

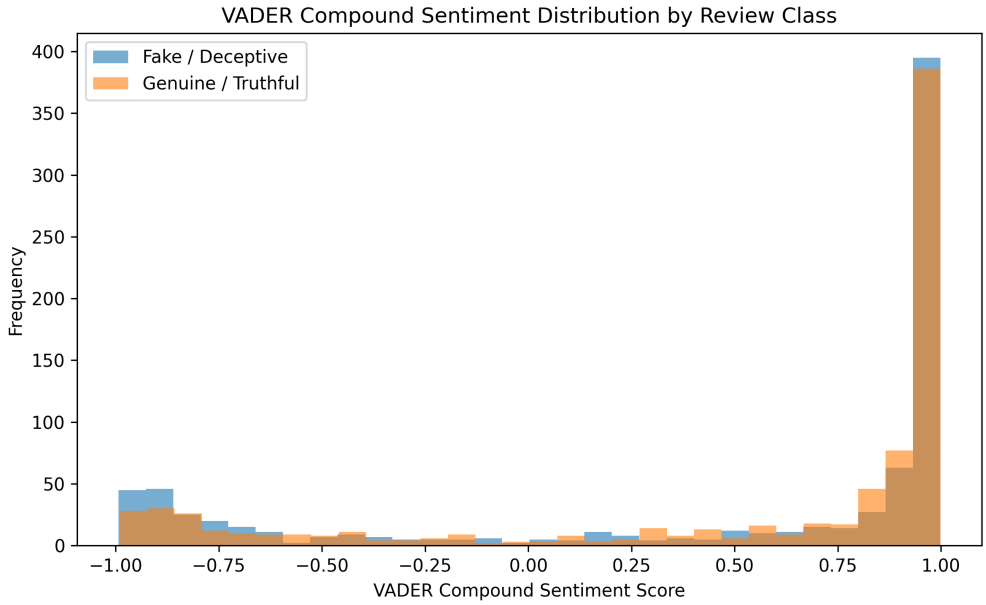


Figure 3.10: VADER Compound Sentiment Distribution by Review Class

Figure 3.10 compares the VADER compound sentiment score distributions between fake and genuine reviews. Deceptive reviews frequently exhibit exaggerated positive sentiment polarity, while genuine reviews reflect more balanced, nuanced evaluation tones.

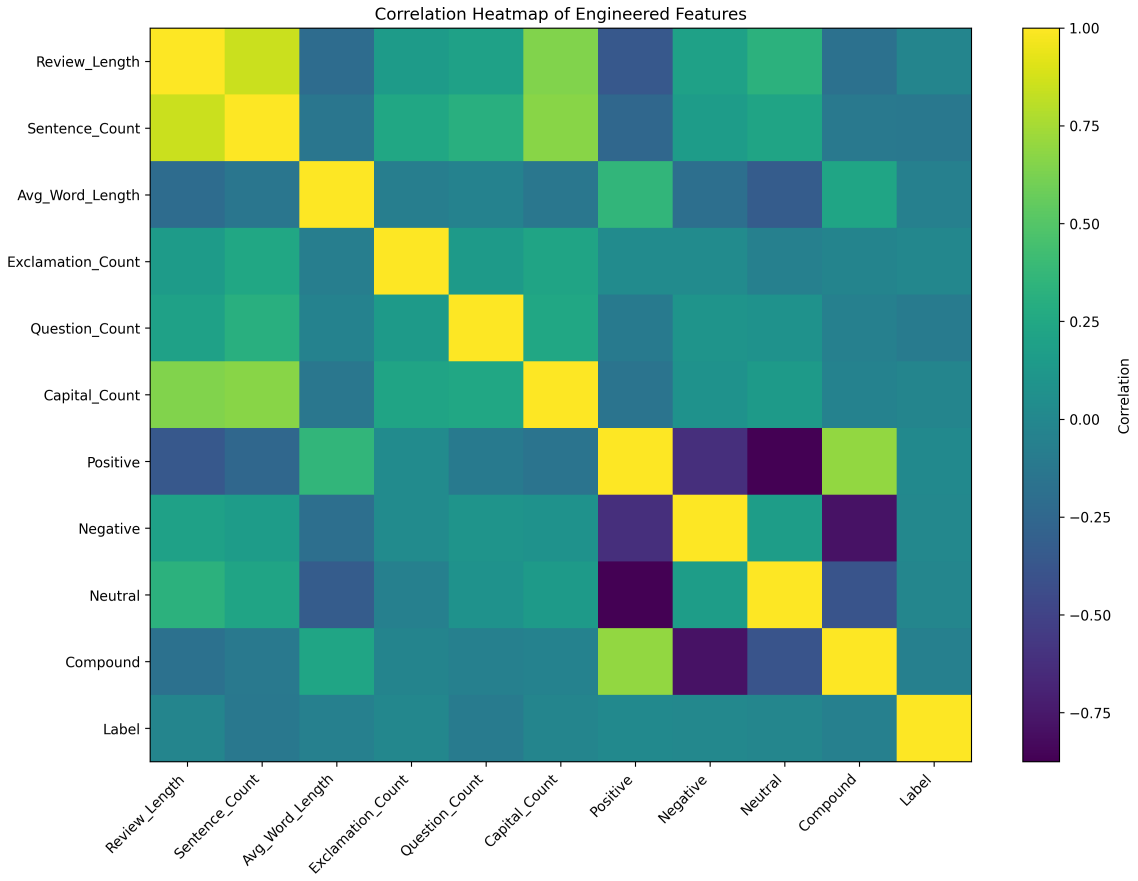


Figure 3.11: Correlation Heatmap of Engineered Features

Figure 3.11 presents the correlation heatmap of the engineered linguistic and sentiment features along with the target class label. The heatmap confirms low inter-feature collinearity while highlighting informative associations with deceptive review patterns.

3.4 Analysis of Algorithms

The proposed Fake Review Detection System uses supervised machine learning algorithms to classify reviews into fake and genuine classes. The input to the classifiers is constructed using four feature configurations: TF-IDF alone, TF-IDF with linguistic features, TF-IDF with VADER sentiment features, and TF-IDF with both linguistic and VADER features. Three classical machine learning algorithms are considered for classification: Logistic Regression, Linear Support Vector Machine (SVM), and Random Forest.

These algorithms are selected because they provide different approaches to binary classification. Logistic Regression provides a probabilistic classification approach, Linear SVM determines a suitable decision boundary by maximizing the separation between classes, and Random Forest uses an ensemble of decision trees to perform classification. The algorithms and feature configurations will be compared using Accuracy, Precision, Recall, F1-score,

ROC–AUC, and Confusion Matrix.

3.4.1 Detailed Study Report of the Proposed Algorithms

Logistic Regression

Logistic Regression is a supervised machine learning algorithm commonly used for binary classification problems. In the proposed system, it will be used to classify a review as either fake or genuine based on the extracted feature representation.

Logistic Regression calculates a weighted combination of the input features and converts the resulting value into a probability using the sigmoid function. The probability is then used to determine the class of the input review.

Working Principle

Let the input feature vector be represented as

$$\mathbf{x} = [x_1, x_2, \ldots, x_n]$$

where each x_i represents one feature of the review. The weighted sum of the features is calculated as

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

where w_i represents the weight associated with a feature and b represents the bias.

The calculated value is passed through the sigmoid function:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-z}}$$

The sigmoid function produces a value between 0 and 1, which can be interpreted as the probability that the review belongs to class 1.

In the proposed system, Label 1 represents a fake or deceptive review and Label 0 represents a genuine or truthful review. Therefore, the classification rule is:

$$P(y = 1|\mathbf{x}) \geq 0.5 \Rightarrow \text{Fake / Deceptive Review}$$

$$P(y = 1|\mathbf{x}) < 0.5 \Rightarrow \text{Genuine / Truthful Review}$$

The model parameters are learned by minimizing the binary cross-entropy loss:

$$L = -[y \log(p) + (1-y) \log(1-p)]$$

where y is the actual class and p is the predicted probability.

During training, the model parameters are optimized to reduce the classification loss. The displayed expression represents the loss for one sample; training minimizes the loss over the training data, with regularization where applicable.

Application in the Proposed System

In the proposed system, Logistic Regression will be examined using different feature configurations based on TF-IDF, linguistic characteristics, and VADER sentiment scores.

The TF-IDF representation consists of unigram and bigram features with a maximum of 10,000 features. Six linguistic features are extracted from the reviews: Review Length, Sentence Count, Average Word Length, Exclamation Count, Question Count, and Capital Count.

In addition, four sentiment-based features are extracted using VADER: Positive, Negative, Neutral, and Compound sentiment scores.

Each feature configuration will be supplied separately to the Logistic Regression classifier for training and comparison. The model will use a regularization parameter of $C = 1.0$ and a maximum of 2000 iterations, as specified in the model-building section. The implementation uses regularized binary classification.

Advantages

- Simple and efficient for binary classification.
- Provides probability-based predictions.
- Works effectively with high-dimensional TF-IDF features.
- Easy to interpret compared with more complex models.
- Requires relatively low computational resources.

Limitations

- Assumes a linear decision boundary between the classes.
- May not capture complex non-linear relationships between features.

- Performance can depend on the quality of the feature representation.

Activity Diagram

The activity diagram for Logistic Regression summarizes the conceptual stages of model training and prediction. The optimization steps represent the learning process rather than a specific solver implementation.

Figure 3.12 illustrates the training and prediction process of Logistic Regression. During training, the model will initialize its parameters, calculate the linear score, apply the sigmoid function, calculate the loss, and update the parameters until convergence. After training, the model will calculate the probability for a new review and classify it as fake or genuine using a threshold of 0.5. In accordance with the dataset labels, Label 1 represents a fake or deceptive review, while Label 0 represents a genuine or truthful review.

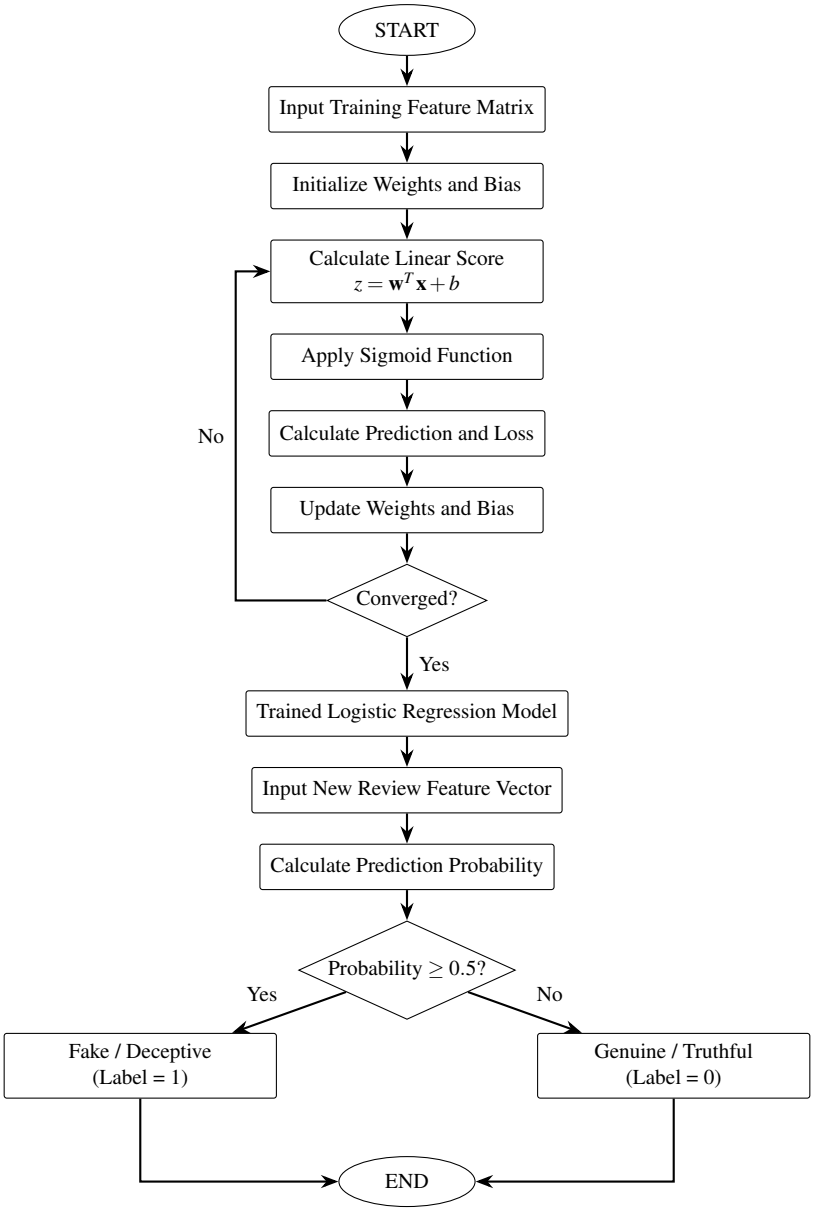


Figure 3.12: Activity Diagram of Logistic Regression

Random Forest

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to perform classification. Instead of depending on a single decision tree, Random Forest creates several decision trees and combines their predictions to obtain the final classification.

The use of multiple trees reduces the dependence on an individual decision tree and allows the model to capture different patterns present in the training data. In the proposed system, Random Forest will be used to classify reviews as either fake/deceptive or genuine/truthful.

Working Principle

Random Forest creates multiple decision trees using different bootstrap samples of the training data. For each tree, a randomly selected subset of features is considered while constructing the tree.

If the forest contains T decision trees, each tree produces a prediction:

$$h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})$$

The final prediction is obtained using majority voting:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})\}$$

The class receiving the highest number of votes from the individual trees becomes the final predicted class.

In the proposed system, Label 1 represents a fake or deceptive review, while Label 0 represents a genuine or truthful review.

Training Process

The training process begins by generating bootstrap samples from the training data. A decision tree is then constructed from each sample. During tree construction, random subsets of features are considered at different nodes. This process is repeated until the required number of trees is created.

In the model-building configuration, the Random Forest classifier will use 300 decision trees. The predictions from these trees will be combined using majority voting to obtain the final classification.

Application in the Proposed System

The Random Forest classifier will be evaluated using the same four feature configurations as the other proposed algorithms, allowing comparison across TF-IDF, linguistic, and sentiment-based representations.

The TF-IDF representation consists of unigram and bigram features with a maximum of 10,000 features. Six linguistic features are extracted from the reviews: Review Length, Sentence Count, Average Word Length, Exclamation Count, Question Count, and Capital Count.

In addition, four sentiment-based features are extracted using VADER: Positive, Negative,

Neutral, and Compound sentiment scores.

Each feature configuration will be provided separately to the Random Forest classifier. The trained trees will combine their predictions to classify a new review as fake/deceptive or genuine/truthful.

Advantages

- Combines multiple decision trees to improve classification stability.
- Can capture non-linear relationships between features.
- Is less sensitive to individual decision tree errors.
- Can handle a large number of input features.
- Provides an ensemble-based classification approach.

Limitations

- Requires more computational resources than simpler linear models.
- A large number of trees can increase training and prediction time.
- The complete ensemble is more difficult to interpret than a single decision tree.

Activity Diagram

The activity diagram for Random Forest represents the process of generating multiple decision trees and combining their predictions to obtain the final class.

Figure 3.13 shows the main training and prediction stages of Random Forest. Bootstrap samples and random feature subsets are used to construct multiple decision trees. After the required number of trees is completed, a new review is passed through the individual trees and their predictions are combined using majority voting to obtain the final classification. In the proposed system, the final classification will represent either a fake/deceptive review (Label 1) or a genuine/truthful review (Label 0).

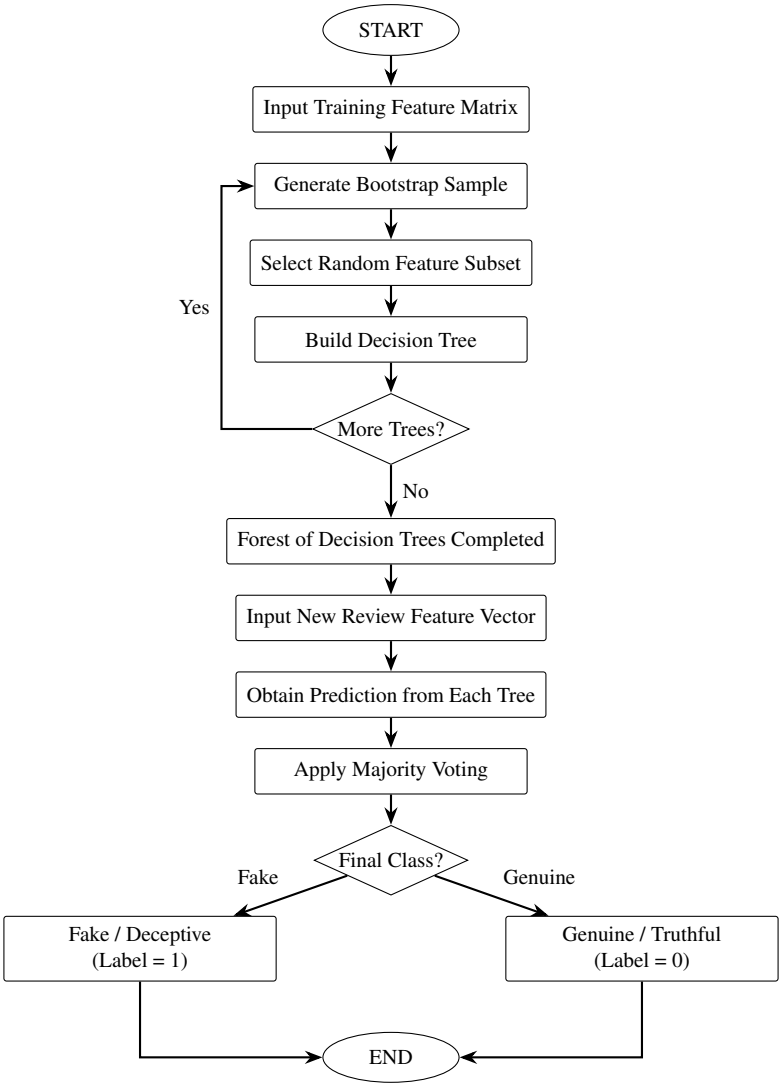


Figure 3.13: Activity Diagram of Random Forest

Linear Support Vector Machine (Linear SVM)

Linear Support Vector Machine (Linear SVM) is a supervised machine learning algorithm used for binary classification. It identifies a linear decision boundary that separates two classes while maximizing the distance between the boundary and the closest training samples.

The proposed system uses Linear SVM because TF-IDF produces a high-dimensional and sparse feature representation, which can also be combined with linguistic or sentiment-based features. Linear SVM is suitable for text classification because it can efficiently operate on high-dimensional and sparse feature representations.

Working Principle

For a linear SVM, the decision boundary is represented by:

$$\mathbf{w}^T\mathbf{x} + b = 0$$

where $\mathbf{w}$ represents the weight vector, $\mathbf{x}$ represents the input feature vector, and b represents the bias.

The decision function is given by:

$$f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$$

The predicted class is determined from the sign of the decision function. In the proposed system, Label 1 represents a fake or deceptive review, while Label 0 represents a genuine or truthful review. Therefore, the classification rule is:

$$f(\mathbf{x}) \geq 0 \Rightarrow \text{Fake / Deceptive Review}$$

$$f(\mathbf{x}) < 0 \Rightarrow \text{Genuine / Truthful Review}$$

The objective of Linear SVM is to identify a separating hyperplane with a large margin while allowing classification errors when necessary. The margin for a linear SVM is related to the magnitude of the weight vector:

$$\text{Margin} = \frac{2}{\|\mathbf{w}\|}$$

For a hard-margin formulation, the optimization problem can be written as:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2$$

subject to:

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1$$

where y_i is the signed class label in $\{-1, +1\}$. This is the mathematical convention used for the margin constraint; the original dataset labels remain 0 (genuine) and 1 (fake) and are handled by the classifier internally.

For practical datasets, Linear SVM uses a soft-margin formulation with regularization parameter C, which balances a large margin against classification errors. The hard-margin equations above are provided to explain the underlying principle.

Support Vectors

Support vectors are the training samples that are closest to the decision boundary. These samples play an important role in determining the position of the separating hyperplane. Changes in the support vectors can affect the position of the decision boundary, whereas samples that are far from the boundary have less direct influence on it.

Application in the Proposed System

In the proposed system, `LinearSVC` will be evaluated using four feature configurations constructed from TF-IDF, linguistic features, and VADER sentiment scores.

The TF-IDF representation consists of unigram and bigram features with a maximum of 10,000 features. Six linguistic features are extracted from the reviews: Review Length, Sentence Count, Average Word Length, Exclamation Count, Question Count, and Capital Count.

In addition, four sentiment-based features are extracted using VADER: Positive, Negative, Neutral, and Compound sentiment scores.

Each feature configuration will be supplied separately to `LinearSVC`. The regularization parameter will be set to $C = 1.0$; the maximum number of iterations will be set to 2000. The trained model will calculate a decision value for an unseen review and use its sign to classify the review as fake/deceptive or genuine/truthful.

Advantages

- Effective for high-dimensional feature spaces.
- Works well with sparse TF-IDF representations.
- Uses a clear linear decision boundary for classification.
- Linear SVM is computationally suitable for text feature spaces.

Limitations

- Does not directly provide probability estimates in the same way as Logistic Regression.
- Performance depends on the selected model parameters and feature representation.
- A linear SVM may not capture complex non-linear relationships between features.

Activity Diagram

The activity diagram for Linear SVM represents the process of learning the separating hyperplane and using it to classify a new review.

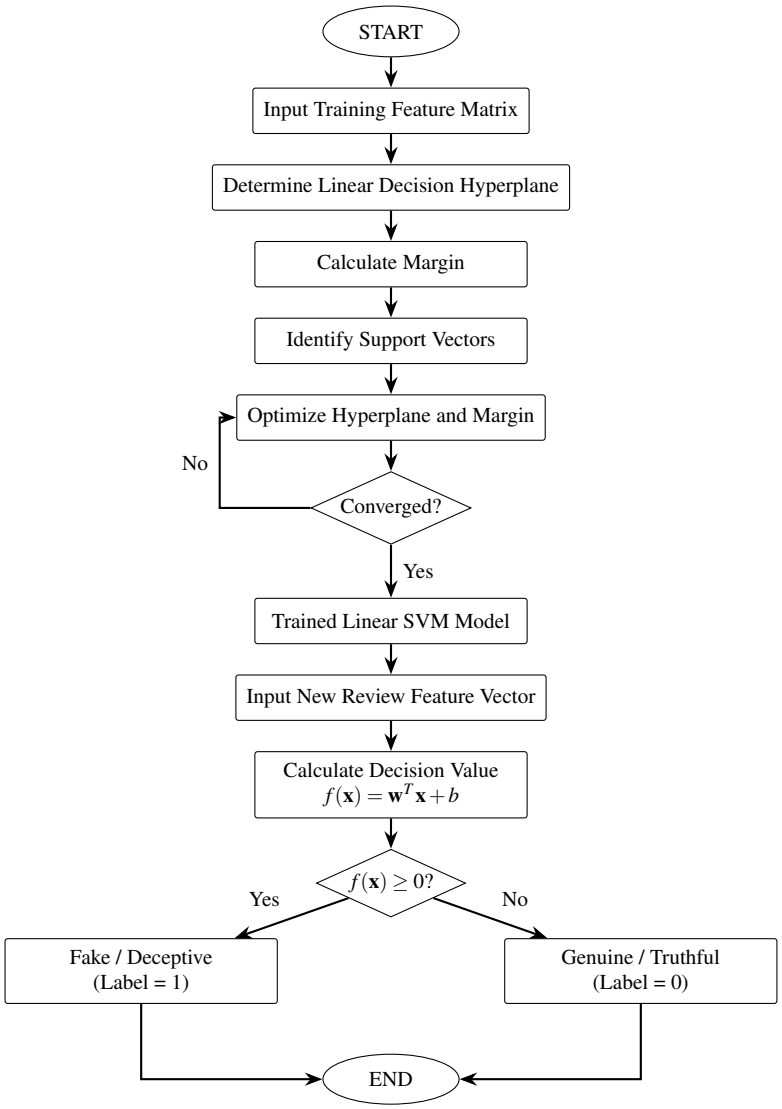


Figure 3.14: Activity Diagram of Linear Support Vector Machine

Figure 3.14 represents the main stages of the Linear SVM algorithm. The model learns a linear separating hyperplane by optimizing a regularized classification objective. The diagram presents the margin and support vectors conceptually. After training, the decision function is calculated for a new review and the sign of the decision value determines the predicted class. In accordance with the dataset labels, a non-negative decision value corresponds to a fake/deceptive review (Label 1), while a negative decision value corresponds to a genuine/truthful review (Label 0).

3.5 Project Pipeline

3.5.1 Project Pipeline Diagram and Development Plan

The proposed Fake Review Detection System follows a structured machine learning pipeline that transforms raw hotel review data into a suitable feature representation for classifying reviews as fake/deceptive or genuine/truthful. The major stages include dataset loading, EDA, preprocessing, feature extraction, feature selection, model training, evaluation, and prediction.

Initially, the Ott Deceptive Opinion Spam Corpus is loaded into the system. It contains hotel review text, ratings, and class labels. The review text is used for feature extraction, while the label is used as the target variable. Label 1 represents fake/deceptive reviews, whereas Label 0 represents genuine/truthful reviews.

Exploratory Data Analysis (EDA) is performed to understand the dataset characteristics, including class distribution and review length. The dataset is also checked for missing values and duplicate records.

The review text is preprocessed using NLP techniques, including HTML tag removal, lowercase conversion, removal of URLs, numbers, punctuation, and special characters, followed by tokenization, English stop-word removal, and lemmatization.

Feature engineering is then performed using TF-IDF with unigram and bigram features, with a maximum of 10,000 features. Six linguistic features are extracted: Review Length, Sentence Count, Average Word Length, Exclamation Count, Question Count, and Capital Count. VADER sentiment analysis provides four additional features: Positive, Negative, Neutral, and Compound scores.

Feature selection is performed to retain relevant features for classification. The dataset is divided using a stratified 80:20 train-test split. The TF-IDF vectorizer is fitted only on the training data and then applied to both training and testing data. The selected linguistic and sentiment features are combined with the TF-IDF representation to form the final feature matrix.

The combined training features are used to train three classifiers: Logistic Regression, Linear SVM, and Random Forest. Logistic Regression is configured with $C = 1.0$ and a maximum of 2000 iterations, Linear SVM uses a maximum of 2000 iterations, and Random Forest uses 300 decision trees.

The trained models are evaluated using Accuracy, Precision, Recall, F1-score, ROC–AUC, and Confusion Matrix. Their performance is compared, and the best-performing model is selected based on the overall evaluation results.

Finally, the selected model is deployed to classify unseen reviews. A new review undergoes

the same preprocessing and feature extraction steps before being supplied to the trained model, which predicts whether it is fake/deceptive (Label 1) or genuine/truthful (Label 0).

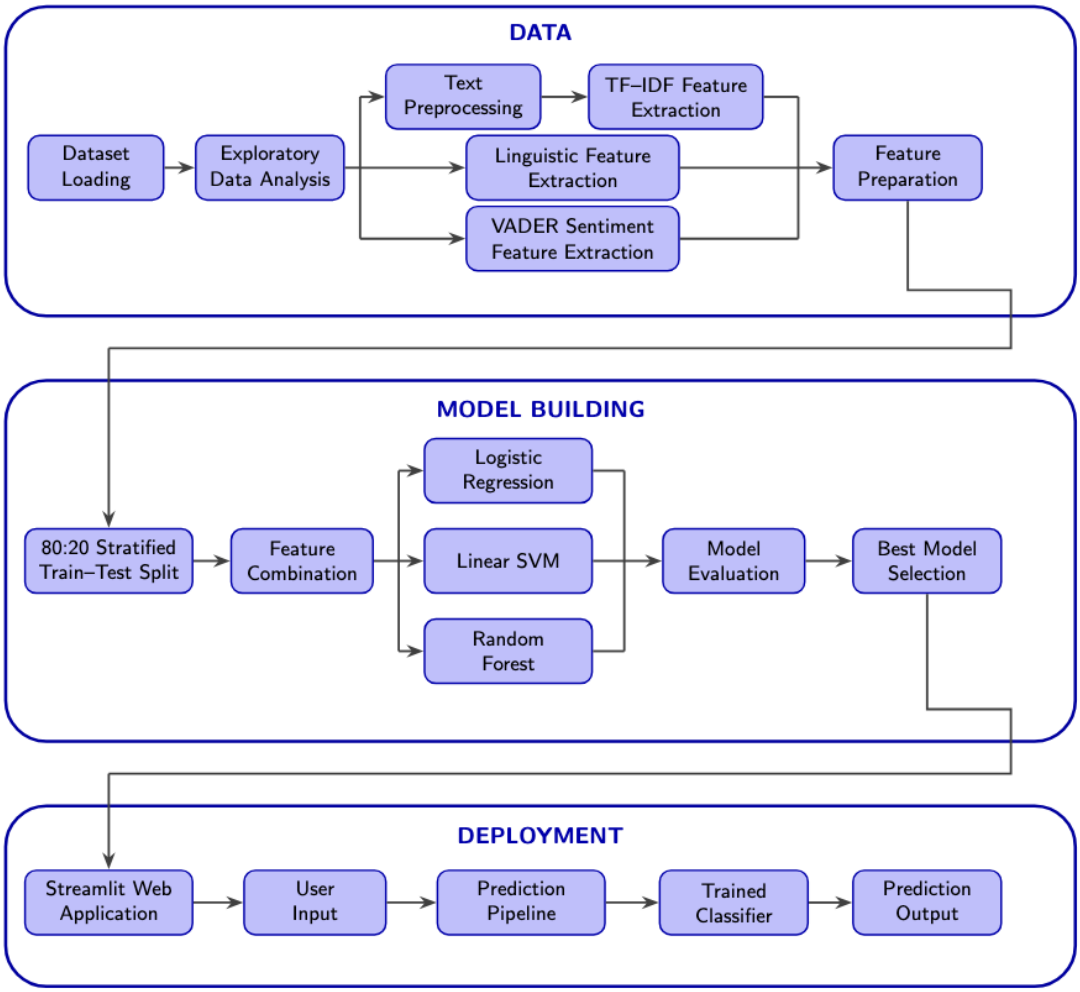


Figure 3.15: Project Pipeline

Project Development Plan and Timeline

The project timeline outlines the major activities planned for the development of the Fake Review Detection System, from dataset analysis and text preprocessing to feature extraction, machine learning model development, evaluation, application implementation, testing, and final submission. The activities are organized according to the common project schedule and presentation milestones.

The planned project activities are:

Period / Date	Planned Activity
Week 1–2	Dataset Collection, Exploratory Data Analysis and Initial Text Preprocessing
21.07.2026	Project Proposal Approval
Week 3	Feature Extraction and Feature Dataset Preparation using TF-IDF, Linguistic and VADER Sentiment Features
Week 4	Feature Analysis, Feature Selection and Visualization
Week 5	Machine Learning Pipeline Preparation and Analysis of Selected Algorithms
Week 6	Finalization of Preprocessing, Feature Engineering and Model Implementation Setup
09.09.2026	First Project Presentation – Sprint Release I
Week 8	Model Training using Logistic Regression, Linear SVM and Random Forest
18.09.2026	Sprint Release II
Week 9	Model Performance Evaluation, Model Comparison and Selection of the Best-Performing Model
Week 10	Web Interface Development, Review Input Interface and CSV Upload/Prediction Implementation
29.09.2026 – 30.09.2026	Interim Project Presentation
Week 11	Prediction Result Interface, Classification Report Generation, System Integration, Testing and Refinement
05.10.2026	Project Execution, Report and PPT Preparation
09.10.2026	Sprint Release III
22.10.2026 – 23.10.2026	Final Project Presentation
30.10.2026	Final Report Submission

Table 3.4: Project Timeline

3.6 Feasibility Analysis

Feasibility analysis evaluates whether the proposed Fake Review Detection System can be successfully developed using the available resources, technologies, and project duration. The system is analysed from technical, operational, economic, and schedule perspectives.

3.6.1 Technical Feasibility

The proposed system is technically feasible as it uses established Natural Language Processing techniques and classical machine learning algorithms. The Ott Deceptive Opinion Spam Corpus provides labelled hotel reviews suitable for supervised binary classification.

Python and commonly available libraries are used for data processing, text preprocessing, feature extraction, sentiment analysis, and machine learning. TF-IDF is used for text representation, while VADER provides sentiment-based features. Linguistic features are also extracted from the reviews. Logistic Regression, Linear SVM, and Random Forest will be implemented using Scikit-learn. NLTK is used for NLP preprocessing and VADER, while Pandas, NumPy, and Matplotlib support data processing and visualization.

Therefore, the required software tools and computational resources are available, making the system technically feasible.

3.6.2 Operational Feasibility

The proposed system is operationally feasible because the review classification process can be automated after model training. A new review can pass through the same preprocessing and feature extraction pipeline before being supplied to the trained classifier.

The system provides a simple classification outcome indicating whether a review is fake/deceptive or genuine/truthful. This can assist in identifying potentially deceptive reviews with minimal manual intervention.

3.6.3 Economic Feasibility

The proposed system is economically feasible because it uses a publicly available dataset, open-source programming language, and freely available machine learning and NLP libraries.

Python, Pandas, NumPy, NLTK, Scikit-learn, and Matplotlib are freely available, eliminating the need for specialised commercial software. Therefore, the overall development cost is low and suitable for an academic project.

3.6.4 Schedule Feasibility

The proposed project is feasible within the available academic project duration. The major stages include literature review, dataset analysis, preprocessing, feature engineering, feature selection, model development, model evaluation, testing, documentation, and final report preparation.

The use of established NLP techniques and classical machine learning algorithms reduces implementation complexity and supports timely completion. Therefore, the proposed Fake

Review Detection System can be completed within the planned project schedule.

3.7 System Environment

The system environment specifies the software tools, libraries, and hardware resources required for the development and implementation of the proposed Fake Review Detection System. The project uses standard computing resources for data processing, text preprocessing, feature extraction, model development, evaluation, and visualization.

3.7.1 Software Environment

The software environment consists of Python and open-source libraries for data processing, Natural Language Processing, machine learning, sentiment analysis, and visualization. Google Colab is used for model development and experimentation, while Visual Studio Code supports application development.

Software / Tool	**Purpose**
Python	Programming language for system implementation.
Google Colab	Model development, training, and experimentation.
Visual Studio Code	Application development and code editing.
Pandas	Dataset handling and data analysis.
NumPy	Numerical computations and array operations.
NLTK	Text preprocessing and sentiment analysis.
Scikit-learn	Machine learning model training and evaluation.
Matplotlib	Data visualization and graphical analysis.
VADER	Extraction of sentiment-based features.
Streamlit	Development of the web-based prediction interface.
Joblib	Saving and loading trained models and feature transformers.

Table 3.5: Software Environment

3.7.2 Hardware Environment

The Fake Review Detection System does not require specialized hardware for development or deployment. A standard laptop or desktop computer with adequate processing power, memory, and storage is sufficient for data processing, model training, evaluation, and application

development.

The basic hardware requirements are presented below.

Hardware Component	**Minimum Requirement**
Processor	Intel Core i5 or equivalent processor.
RAM	8 GB or higher.
Storage	256 GB or higher.
Display	Standard laptop or desktop display.
Internet Connection	Stable internet connection for downloading datasets, libraries, and accessing online resources.

Table 3.6: Hardware Environment

4. System Design

4.1 Model Building

The Fake Review Detection system was designed to classify reviews as fake/deceptive or genuine/truthful using information extracted directly from the review text. Three types of information were considered: textual, linguistic, and sentiment-based features.

TF-IDF (Term Frequency-Inverse Document Frequency) was used to represent important words and phrases in the reviews. Six linguistic features were extracted to represent writing characteristics: Review Length, Sentence Count, Average Word Length, Exclamation Count, Question Count, and Capital Count. Sentiment information was obtained using VADER through four scores: Positive, Negative, Neutral, and Compound.

To examine the contribution of these features, four feature configurations were constructed:

1. TF-IDF
2. TF-IDF + Linguistic Features
3. TF-IDF + VADER Features
4. TF-IDF + Linguistic Features + VADER Features

Three supervised machine-learning classifiers were used: Logistic Regression, Linear SVM, and Random Forest. Each classifier was applied to all four feature configurations, resulting in twelve experimental combinations.

This design allows the performance of the three classifiers to be compared while also examining whether linguistic and sentiment information provides additional predictive value beyond the TF-IDF baseline. The models are evaluated using Accuracy, Precision, Recall, F1-score, ROC–AUC, and Confusion Matrix.

4.1.1 Implementation Code

The system was implemented in Python using Google Colab. The implementation consists of text preprocessing, feature extraction, feature transformation, model training, testing, and evaluation.

The original review text in the `Ori_Review` column is preprocessed by converting it to lowercase, removing HTML tags, URLs, numbers, punctuation, and special characters, followed by tokenization, stop-word removal, and lemmatization. The resulting text is stored in `Clean_Review`.

TF-IDF is then applied using unigrams and bigrams with a maximum of 10,000 features. The vectorizer is fitted only on the training data and subsequently used to transform the test data.

Six linguistic features are extracted from the original review text: Review Length, Sentence Count, Average Word Length, Exclamation Count, Question Count, and Capital Count. VADER is used to extract four sentiment features: Positive, Negative, Neutral, and Compound.

The linguistic and VADER features are standardized separately using `StandardScaler`, with the scalers fitted only on the training data. The transformed numerical features are then combined with the TF-IDF representation to construct the required feature configurations.

The four feature configurations are supplied independently to Logistic Regression, Linear SVM, and Random Forest. The resulting models are trained and tested using the same data split, and their predictions are retained for performance evaluation and comparison.

4.1.2 Model Planning (Training / Test Split)

The dataset contains 1,600 labelled hotel reviews, consisting of 800 fake/deceptive reviews (Label 1) and 800 genuine/truthful reviews (Label 0). An 80:20 stratified train–test split is used, resulting in 1,280 training reviews and 320 testing reviews.

Since stratification preserves the balanced class distribution, the training set contains 640 reviews from each class, while the test set contains 160 reviews from each class. The split is performed using `test_size=0.20`, `random_state=42`, and `stratify=y`. The fixed random state supports reproducibility of the experiment.

The same training and testing indices are maintained across all four feature configurations and three classifiers to ensure a consistent comparison. To prevent information leakage, the TF-IDF vectorizer and the scalers for the linguistic and VADER features are fitted only on the training data and subsequently applied to the test data.

4.1.3 Training

Three supervised machine-learning classifiers are trained using the 1,280 training reviews: Logistic Regression, Linear SVM, and Random Forest. Each classifier is trained separately on the four feature configurations, resulting in twelve model–feature-set combinations.

Logistic Regression is configured with `C=1.0`, `max_iter=2000`, and `random_state=42`. Linear SVM uses `C=1.0` and `random_state=42`. Random Forest uses 300 decision trees with `max_depth=None`, `min_samples_split=2`, `random_state=42`, and `n_jobs=-1`.

The same model configurations are maintained across all feature sets so that performance differences can be examined in relation to the feature representation rather than changes in model parameters. The classifier configurations are shown in Figure 4.1.

```
def make_models():
    return {
        "Logistic Regression": LogisticRegression(
            max_iter=2000,
            C=1.0,
            random_state=42
        ),

        "Linear SVM": LinearSVC(
            C=1.0,
            random_state=42
        ),

        "Random Forest": RandomForestClassifier(
            n_estimators=300,
            max_depth=None,
            min_samples_split=2,
            random_state=42,
            n_jobs=-1
        )
    }
```

Figure 4.1: Configuration of Machine-Learning Models Used for Training

For each feature configuration, the classifiers are fitted using the corresponding training feature matrix and training labels. The trained models are then stored together with their model and feature-set names for subsequent testing.

The implementation of the training process is shown in Figure 4.2.

```
models = make_models()

for model_name, model in models.items():

    model.fit(
        X_train_current,
        y_train
    )

    trained_models[
        (feature_name, model_name)
    ] = model
```

Figure 4.2: Training of Machine-Learning Models

4.1.4 Testing (Performance Evaluation on Test Data)

The twelve trained model–feature-set combinations are tested using the 320 unseen test reviews. The same test samples are used for every experiment to provide a consistent comparison between the models and feature representations.

For each trained model, the `predict()` method is applied to the corresponding test feature matrix to generate predicted class labels. Logistic Regression and Random Forest use `predict_proba()` to obtain probability scores, while Linear SVM uses `decision_function()` because `LinearSVC` does not directly provide class probabilities.

The implementation used to generate predictions and scores is shown in Figure 4.3.

```
y_pred = model.predict(
    X_test_current
)

if hasattr(model, "predict_proba"):
    y_score = model.predict_proba(
        X_test_current
    )[:, 1]
else:
    y_score = model.decision_function(
        X_test_current
    )
```

Figure 4.3: Prediction of Class Labels and Scores on the Test Dataset

The predicted labels and scores are retained for performance evaluation using Accuracy, Precision, Recall, F1-score, ROC–AUC, and the Confusion Matrix.

5. Results and Discussion

This chapter presents and discusses the experimental results obtained from the Fake Review Detection system. The evaluation was performed using the unseen test dataset containing 320 reviews, consisting of 160 fake/deceptive reviews and 160 genuine/truthful reviews.

Three machine-learning classifiers—Logistic Regression, Linear SVM, and Random Forest—were evaluated using four feature configurations: TF-IDF, TF-IDF + Linguistic Features, TF-IDF + VADER Features, and TF-IDF + Linguistic + VADER Features. This resulted in twelve model–feature-set combinations.

The analysis focuses on the performance of the classifiers and the contribution of linguistic and sentiment-based information beyond the TF-IDF baseline. Accuracy, Precision, Recall, F1-score, ROC–AUC, and the Confusion Matrix were used for evaluation.

5.1 Model Performance and Evaluation

The twelve experimental configurations were evaluated using Accuracy, Precision, Recall, F1-score, and ROC–AUC. These measures provide different perspectives on the ability of the models to distinguish fake reviews from genuine reviews.

Fake/deceptive reviews are represented by Label 1 and genuine/truthful reviews by Label 0. Therefore, True Positive (TP) represents a fake review correctly classified as fake, True Negative (TN) represents a genuine review correctly classified as genuine, False Positive (FP) represents a genuine review incorrectly classified as fake, and False Negative (FN) represents a fake review incorrectly classified as genuine.

Accuracy

Accuracy represents the overall proportion of correctly classified reviews and is calculated as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{5.1}$$

Precision

Precision represents the proportion of reviews predicted as fake that are actually fake:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5.2}$$

Recall

Recall represents the proportion of actual fake reviews that are correctly identified:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{5.3}$$

F1-score

F1-score represents the harmonic mean of Precision and Recall:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5.4}$$

ROC–AUC

ROC–AUC measures the ability of a classifier to discriminate between fake and genuine reviews across different classification thresholds. For Logistic Regression and Random Forest, probability scores were used, whereas the scores generated by the `decision_function()` were used for Linear SVM.

Figure 5.1 presents the performance obtained from all twelve model–feature-set combinations.

	Feature Set	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	TF-IDF	Logistic Regression	0.9031	0.8957	0.9125	0.9040	0.9682
1	TF-IDF	Linear SVM	0.9094	0.9119	0.9062	0.9091	0.9691
2	TF-IDF	Random Forest	0.8969	0.8994	0.8938	0.8966	0.9607
3	TF-IDF + Linguistic	Logistic Regression	0.8438	0.8125	0.8938	0.8512	0.9221
4	TF-IDF + Linguistic	Linear SVM	0.8969	0.8896	0.9062	0.8978	0.9604
5	TF-IDF + Linguistic	Random Forest	0.8938	0.8889	0.9000	0.8944	0.9551
6	TF-IDF + VADER	Logistic Regression	0.9188	0.9136	0.9250	0.9193	0.9681
7	TF-IDF + VADER	Linear SVM	0.9156	0.9182	0.9125	0.9154	0.9709
8	TF-IDF + VADER	Random Forest	0.8906	0.8882	0.8938	0.8910	0.9572
9	TF-IDF + Linguistic + VADER	Logistic Regression	0.8656	0.8421	0.9000	0.8701	0.9221
10	TF-IDF + Linguistic + VADER	Linear SVM	0.9125	0.9024	0.9250	0.9136	0.9615
11	TF-IDF + Linguistic + VADER	Random Forest	0.8781	0.8712	0.8875	0.8793	0.9505

Figure 5.1: Performance Comparison of Model–Feature-Set Combinations

TF-IDF alone provided a strong baseline for fake-review classification. Logistic Regression achieved an accuracy of 90.31%, Linear SVM achieved 90.94%, and Random Forest achieved 89.69%.

Among all twelve experimental configurations, Logistic Regression with TF-IDF + VADER achieved the highest Accuracy of 91.88% and the highest F1-score of 91.93%. Linear SVM with TF-IDF + VADER achieved the highest Precision of 91.82% and the highest ROC–AUC of 97.09%. The highest Recall of 92.50% was obtained by Logistic Regression with TF-IDF +

VADER and Linear SVM with TF-IDF + Linguistic + VADER.

These results indicate that the different model–feature combinations provide different advantages. Therefore, the performance of the models is analysed using multiple evaluation measures rather than accuracy alone.

5.2 Feature Contribution Analysis

The feature-contribution analysis examines whether linguistic and sentiment-based information provides additional predictive value beyond the TF-IDF textual representation.

TF-IDF Baseline

TF-IDF alone produced strong classification results for all three models. Logistic Regression achieved 90.31% accuracy, Linear SVM achieved 90.94%, and Random Forest achieved 89.69%. This indicates that lexical and phrase-level information within the review text contains substantial predictive information.

Contribution of Linguistic Features

The contribution of the six linguistic features was examined by comparing TF-IDF with TF-IDF + Linguistic Features.

For Logistic Regression, accuracy decreased from 90.31% to 84.38%, a reduction of 5.93 percentage points. Linear SVM decreased from 90.94% to 89.69%, a reduction of 1.25 percentage points, while Random Forest decreased from 89.69% to 89.38%, a reduction of 0.31 percentage points.

Therefore, the selected linguistic features did not improve classification accuracy over the TF-IDF baseline for any of the three classifiers under the current experimental setup. This does not indicate that linguistic information is generally unhelpful; rather, the six selected features did not provide additional predictive value in the present experiment. Although the reviewed literature demonstrates that linguistic features can improve fake review detection, the present experiment produced different results. This variation may be associated with differences in dataset characteristics and feature representations. Previous studies examined different review datasets and broader combinations of linguistic, semantic, and behavioural features, whereas the present study used six structural linguistic features extracted from hotel reviews.

Additionally, TF-IDF already captures important textual patterns, and the selected linguistic features may not have provided sufficient complementary information to improve classification performance. The observed decrease in accuracy does not establish that linguistic features are ineffective in general. Instead, it highlights that the usefulness of additional features depends on the dataset, selected characteristics, and classification algorithm.

Contribution of VADER Sentiment Features

The contribution of sentiment information was examined by comparing TF-IDF with TF-IDF + VADER Features.

For Logistic Regression, accuracy increased from 90.31% to 91.88%. Precision increased from 89.57% to 91.36%, Recall increased from 91.25% to 92.50%, and F1-score increased from 90.40% to 91.93%. ROC–AUC remained almost unchanged, changing from 96.82% to 96.81%.

For Linear SVM, accuracy increased from 90.94% to 91.56%. Precision increased from 91.19% to 91.82%, Recall increased from 90.62% to 91.25%, F1-score increased from 90.91% to 91.54%, and ROC–AUC increased from 96.91% to 97.09%.

Random Forest showed a different pattern. Its accuracy decreased from 89.69% to 89.06%, while its F1-score decreased from 89.66% to 89.10%.

These results indicate that the contribution of VADER sentiment information is model-dependent. It provided additional predictive value for Logistic Regression and Linear SVM, while the same improvement was not observed for Random Forest.

Combined Feature Contribution

The complete feature representation combines TF-IDF, linguistic features, and VADER sentiment features.

Using this representation, Logistic Regression achieved 86.56% accuracy, Linear SVM achieved 91.25%, and Random Forest achieved 87.81%.

Compared with TF-IDF + VADER, adding the linguistic features reduced accuracy for all three classifiers. Therefore, combining all available features did not produce the highest classification performance.

This demonstrates that increasing the number of features does not necessarily improve classification performance. The usefulness of additional features depends on the information they provide and how the learning algorithm uses that information.

Accuracy Comparison

Figure 5.2 presents the accuracy obtained by the three classifiers across the four feature configurations.

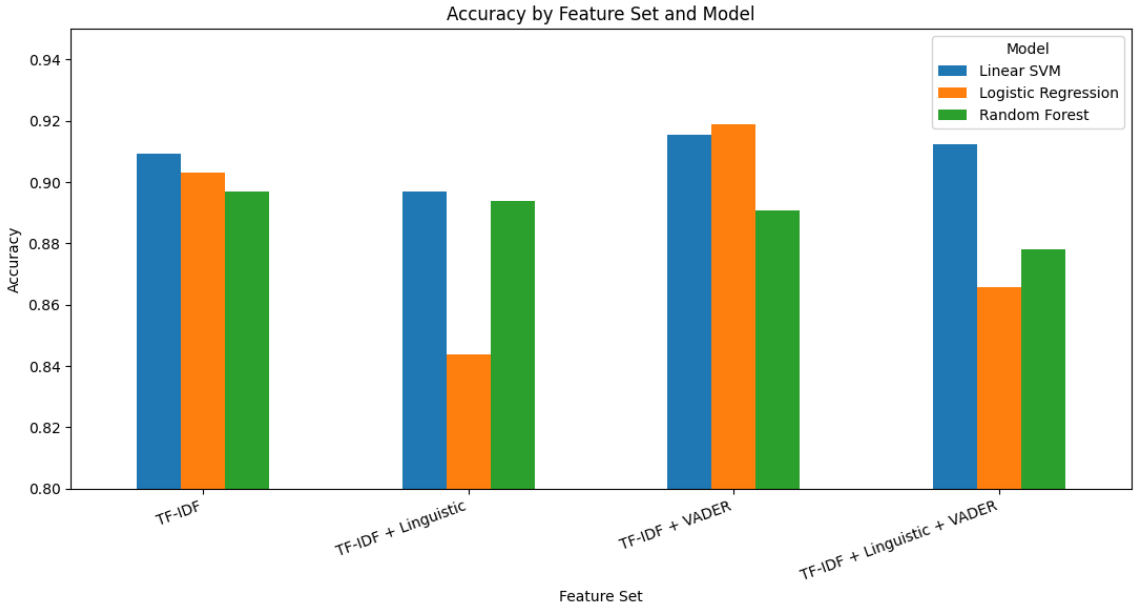


Figure 5.2: Accuracy Comparison Across Feature Sets and Classification Models

The accuracy comparison clearly shows that TF-IDF provides a strong baseline. The addition of VADER features improves Logistic Regression and Linear SVM, whereas the selected linguistic features do not produce an accuracy improvement over the baseline.

F1-Score Comparison

Figure 5.3 presents the F1-score obtained across the feature configurations.

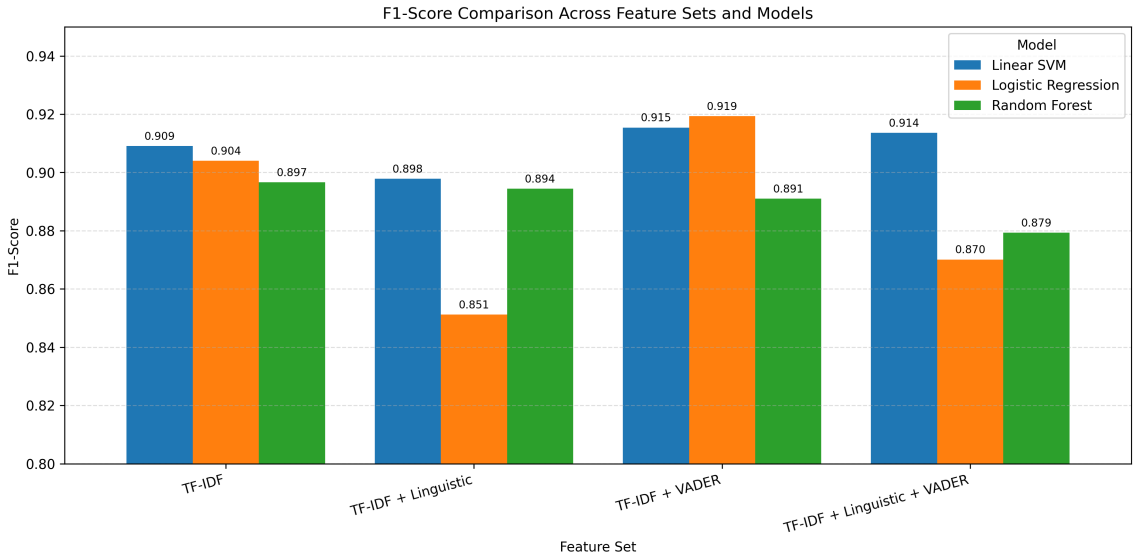


Figure 5.3: F1-score Comparison Across Feature Sets and Classification Models

The F1-score comparison follows a similar pattern to the accuracy results. Logistic Regression with TF-IDF + VADER achieved the highest F1-score of 91.93%.

5.3 ROC–AUC Analysis

ROC–AUC was used to evaluate the ability of the classifiers to discriminate between fake/deceptive and genuine/truthful reviews across different classification thresholds.

The Receiver Operating Characteristic (ROC) curve represents the relationship between the True Positive Rate and False Positive Rate. A larger area under the ROC curve indicates stronger discrimination between the two classes.

Figure 5.4 presents the ROC curves obtained using the TF-IDF + VADER feature configuration for the three classifiers.

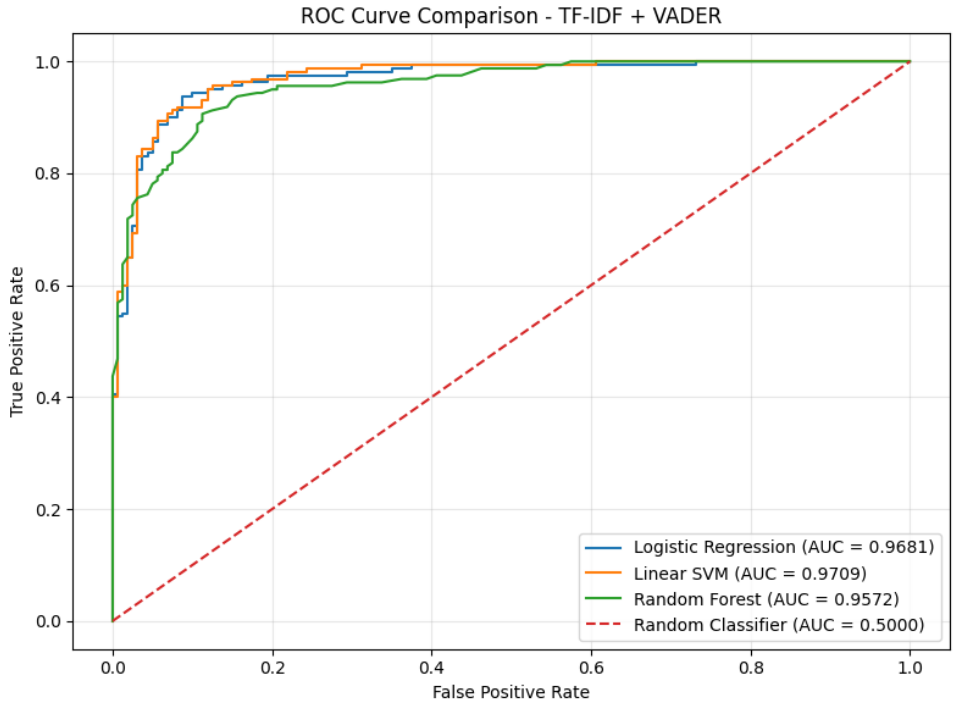


Figure 5.4: ROC Curve Comparison for TF-IDF + VADER Models

For the TF-IDF + VADER configuration, Linear SVM achieved the highest ROC–AUC value of 0.9709, followed by Logistic Regression with 0.9681 and Random Forest with 0.9572.

Although Linear SVM achieved the highest ROC–AUC, Logistic Regression with TF-IDF + VADER achieved the highest Accuracy and F1-score. Therefore, multiple evaluation measures were considered when analysing the overall model performance.

5.4 Confusion Matrix Analysis

Logistic Regression with TF-IDF + VADER was selected for detailed analysis because it achieved the highest Accuracy and F1-score among the twelve evaluated configurations.

The confusion matrix provides a detailed representation of correct and incorrect classifications.

True Positive represents a fake review correctly classified as fake, while True Negative represents a genuine review correctly classified as genuine. False Positive represents a genuine review incorrectly classified as fake, whereas False Negative represents a fake review incorrectly classified as genuine.

Figure 5.5 shows the confusion matrix obtained for Logistic Regression with TF-IDF + VADER.

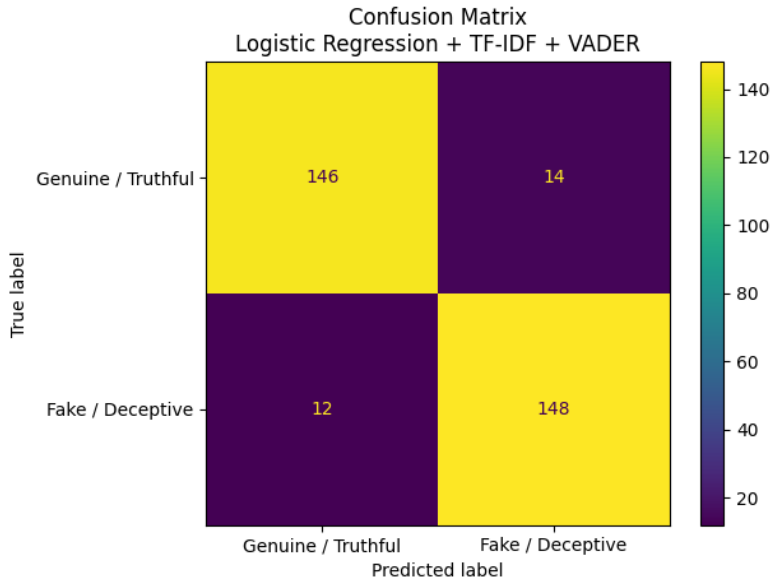


Figure 5.5: Confusion Matrix of Logistic Regression with TF-IDF + VADER

The diagonal elements of the confusion matrix represent correctly classified reviews, while the off-diagonal elements represent classification errors.

Table 5.1 summarizes the evaluation results obtained by the selected configuration.

Evaluation Metric	**Result**
Accuracy	91.88%
Precision	91.36%
Recall	92.50%
F1-score	91.93%
ROC–AUC	96.81%

Table 5.1: Performance of Logistic Regression with TF-IDF + VADER

The Accuracy of 91.88% indicates that a high proportion of the test reviews were correctly classified. Precision of 91.36% indicates that most reviews predicted as fake were actually fake, while Recall of 92.50% indicates that the model identified a high proportion of the actual fake reviews.

The F1-score of 91.93% demonstrates a strong balance between Precision and Recall. The

ROC–AUC value of 96.81% further indicates strong discrimination between fake and genuine reviews.

5.5 Prediction and Testing on Unseen Reviews

To examine the performance of the developed Fake Review Detection System on previously unseen data, an external labelled dataset of hotel reviews without overlap with the training corpus was considered.

To evaluate the system under a balanced test configuration comparable to the training class distribution, a balanced evaluation dataset of 500 reviews was created through stratified random sampling. The evaluation set contained an equal distribution of 250 Fake/Deceptive reviews and 250 Genuine/Truthful reviews.

The 500 unseen reviews were provided to the already-trained Logistic Regression model without retraining or modifying the existing prediction pipeline. Each review was processed using the same preprocessing and feature extraction procedures applied during model development. The cleaned review text was transformed using the fitted TF-IDF vectorizer, while VADER sentiment features were extracted from the original review text and standardized using the fitted scaler.

The resulting TF-IDF and VADER features were combined and supplied to the trained Logistic Regression classifier. The predicted labels were compared with the actual ground-truth labels to determine whether each review was correctly classified.

Out of the 500 unseen reviews tested, the model correctly classified 452 reviews, while 48 reviews were incorrectly classified. These results indicate that the model maintained consistent classification ability on independent, unseen review data.

Description	**Number of Reviews**
Total Balanced Reviews Tested	500
Fake / Deceptive Reviews Tested	250
Genuine / Truthful Reviews Tested	250
Correctly Classified Reviews	452
Incorrectly Classified Reviews	48

Table 5.2: Summary of External Dataset Testing

5.5.1 Example Review Predictions

To demonstrate the practical operation of the prediction system, four reviews from the external dataset were individually entered into the Streamlit web application.

The system processed each review and displayed the predicted class together with the estimated fake and genuine probabilities.

The following examples illustrate the prediction results obtained through the developed application.

Review 1

> We booked a room at the Hilton Chicago for two nights to stay the weekend but upon arrival our reservation could not be found. I even provided the hotel with our confirmation number but it still wasn't available! Fortunately another reservation had not shown up so there was space available for us. Once we settled into our room for the night (after a long flight), there was a bad odor coming from the bathroom. Apparently the cleaning solution they used was tainted and they had to re-clean it after our arrival. Needless to say, with both of these poor experiences in just one night, I will be hesitant to EVER stay in this hotel again, let alone any other Hilton around. Please consider someplace else before the Hilton Chicago!

Predicted Class: Fake / Deceptive Review
Fake Probability: 72.69%
Genuine Probability: 27.31%

Review 2

> Excellent location, feels like a boutique hotel. Right next to Neiman Marcus (V.expensive dept store) and 2 mins walk to John Hancock tower. As most American medium to high end hotels service was excellent from the bellhop to reception. Upgrade to a suite if you are staying longer than a couple of days as walking around the windy city can be tiring and a suite feels like a small apartment. The pub in the basement is good for a sandwich and a Beer. They will be refurbing so take adavantage of lower rates now as I feel they will be going for very high end clients after the remodelling.

Predicted Class: Genuine / Truthful Review
Fake Probability: 27.51%
Genuine Probability: 72.49%

Review 3

> Extremely disappointed in our stay. This hotel did not look like what pictures portrayed online. I booked online thinking what a perfect place this was going to be, only to arrive and have a room without a view even though that is what was paid for. The room smelled musty and not clean. Garbage was not changed from prior person. Found small ants in bathroom near shower. Called to complain and someone came up 2 hours later and used bug spray. Overall this hotel is in a great location, but we would never visit again. Staff was not overly friendly or helpful and the room was not as all as it seemed online.

Predicted Class: Fake / Deceptive Review
Fake Probability: 53.38%
Genuine Probability: 46.62%

Review 4

> My husband & I stayed at the Fitzpatrick in early June 2004 for my birthday-great hotel! Location provides easy walking to Navy Pier, Marshall Field, Michigan Avenue & John Hancock. Room seemed spacious even though its only about 300 sq ft. Lots of room in bathroom; comfortable bed; very quiet, upscale hotel. We would definitely stay here again!

Predicted Class: Genuine / Truthful Review
Fake Probability: 23.69%
Genuine Probability: 76.31%

Prediction Summary

No.	Review Summary	Predicted Class	Fake Probability	Genuine Probability
1	Reservation difficulties, unpleasant bathroom odor, and negative hotel experience	Fake / Deceptive	72.69%	27.31%
2	Positive description of hotel location, service, room upgrade, and facilities	Genuine / Truthful	27.51%	72.49%
3	Negative experience involving room condition, cleanliness, insects, and poor service	Fake / Deceptive	53.38%	46.62%
4	Positive experience describing hotel location, room facilities, and willingness to return	Genuine / Truthful	23.69%	76.31%

Table 5.3: Prediction Results for Selected External Reviews

5.5.2 Inference from External Testing

The external testing demonstrated that the trained Logistic Regression model was able to classify a substantial number of previously unseen hotel reviews without additional training.

Among the 500 balanced unseen reviews tested, 452 were classified correctly and 48 were classified incorrectly. This indicates that the model was able to recognize patterns associated with deceptive and truthful reviews in a separate, unseen dataset.

The four selected examples further demonstrate the practical functionality of the Streamlit application. Two reviews were predicted as Fake/Deceptive and two were predicted as Genuine/Truthful.

The estimated fake probabilities varied across the examples, reflecting differences in the model's confidence. The third review produced a fake probability of 53.38%, indicating a prediction close to the classification threshold.

Since these examples were selected from the labelled external dataset, their predictions can be compared with the corresponding actual labels to determine individual correctness.

The results also show that the system does not classify every review correctly. Differences in writing style, vocabulary, and sentiment expression may influence the predictions.

Overall, the external testing supports the practical applicability of the developed system for identifying potentially deceptive hotel reviews. However, the predicted classification should be treated as a machine learning assessment rather than definitive proof of review authenticity.

5.6 Discussion of Findings

The experimental results provide several observations regarding the use of review-text information for fake-review detection.

First, TF-IDF alone produced strong classification performance across all three algorithms, with accuracies ranging from 89.69% to 90.94%. This indicates that lexical and phrase-level information within review text contains substantial predictive information.

Second, the selected linguistic features did not improve classification accuracy when added to TF-IDF. The reduction was largest for Logistic Regression and smaller for Linear SVM and Random Forest, indicating that these features did not provide additional predictive value under the present experimental setup.

Third, VADER sentiment features improved the two linear classifiers. Logistic Regression improved from 90.31% to 91.88% accuracy, while Linear SVM improved from 90.94% to 91.56%. Random Forest did not show the same improvement, indicating that the contribution of sentiment information depends on the learning algorithm.

The complete TF-IDF + Linguistic + VADER representation did not produce the strongest overall performance, showing that increasing the number of features does not necessarily improve classification.

Among the evaluated configurations, Logistic Regression with TF-IDF + VADER achieved the highest Accuracy of 91.88% and F1-score of 91.93%, while Linear SVM with TF-IDF + VADER achieved the highest Precision of 91.82% and ROC–AUC of 97.09%.

Logistic Regression with TF-IDF + VADER was selected for the final prediction pipeline based on its Accuracy and F1-score performance and its ability to provide probability estimates through `predict_proba()`.

Overall, TF-IDF provided a strong textual baseline, while VADER sentiment information provided additional value for Logistic Regression and Linear SVM. The selected linguistic features did not improve performance over the TF-IDF baseline in this experiment.

6. Model Deployment

The proposed Fake Review Detection system is deployed using a Streamlit-based web application that integrates the trained machine learning model for real-time classification of hotel reviews. The selected Logistic Regression model, trained using TF-IDF and VADER sentiment features, is stored in serialized form (`logistic_regression_model.pkl`) and loaded at runtime using Joblib. The fitted TF-IDF vectorizer and VADER scaler are also saved and loaded to ensure consistency between the training and deployment environments.

When a user accesses the application, a text input interface allows them to enter a hotel review for analysis. The submitted review undergoes input validation, followed by text preprocessing operations such as lowercase conversion, removal of HTML tags, URLs, numbers and punctuation, tokenization, stop-word removal, and lemmatization. The cleaned text is transformed into numerical features using the saved TF-IDF vectorizer. Simultaneously, VADER sentiment analysis is performed on the original review to extract positive, negative, neutral, and compound sentiment scores. These sentiment features are standardized using the saved VADER scaler.

Once feature extraction is complete, the TF-IDF and scaled VADER features are combined into a single feature vector and passed to the trained Logistic Regression classifier. The model classifies the review as either Fake (Deceptive) or Genuine (Truthful) and generates the corresponding prediction probabilities. The classification result is displayed through the Streamlit interface in a clear and user-friendly manner.

To improve transparency, the application also provides a detailed explanation of the prediction process. This includes the preprocessed review text, important TF-IDF terms, extracted VADER sentiment scores, and the contributions of individual features to the Logistic Regression decision score. The application also incorporates input validation and error-handling mechanisms to manage empty, insufficient, or unrecognized review inputs and prevent invalid submissions from proceeding to classification.

6.1 Model Implementation and User Interface Design

The deployed Fake Review Detection system provides an interactive web interface through which users can submit hotel reviews and obtain classification results. The application also presents the main processing stages involved in generating the prediction, improving the transparency and interpretability of the proposed system. The following screenshots illustrate the user interface and the prediction explanation provided by the deployed application.

Figure 6.1 shows the main interface of the deployed application. The user can enter a hotel review and click the *Analyze Review* button to obtain a prediction. The application displays the predicted class as either Fake/Deceptive or Genuine/Truthful, along with the corresponding prediction probabilities.

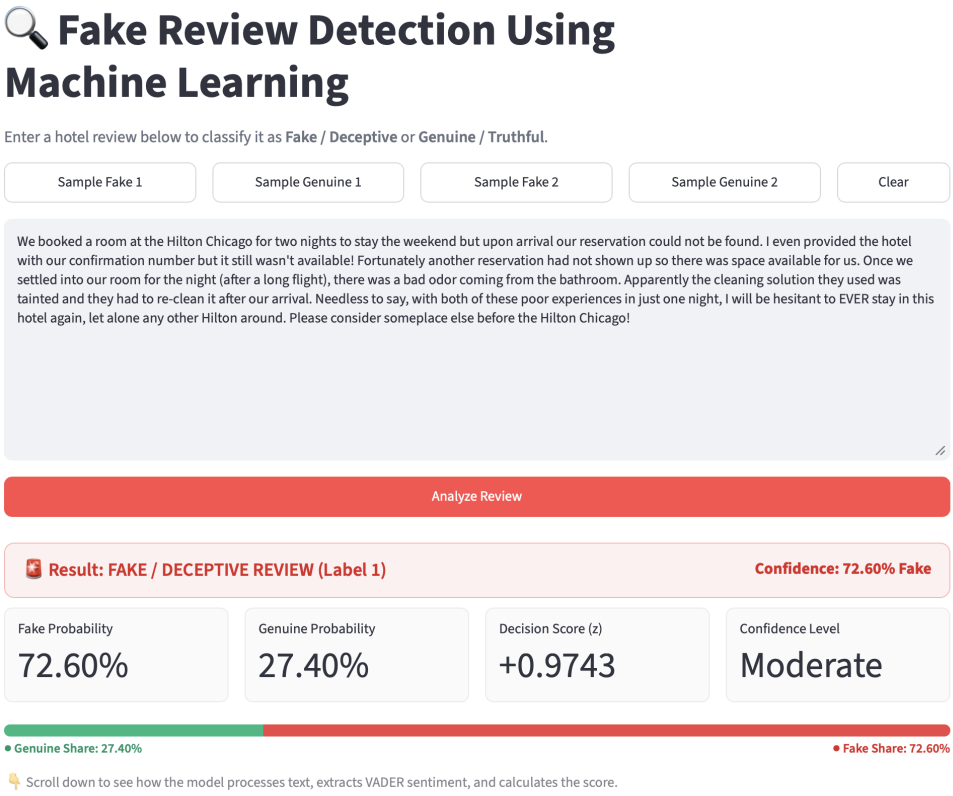


Figure 6.1: Hotel review classification result and prediction probabilities.

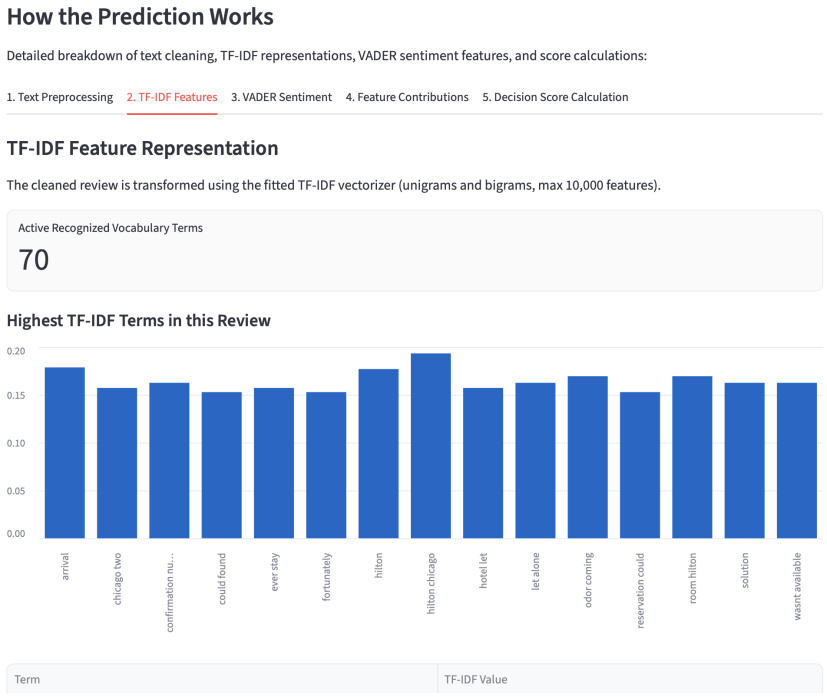


Figure 6.2: Explanation of the prediction process and Logistic Regression decision score.

7. Git History

The Git repository for the Fake Review Detection Using Machine Learning project contains the essential components required for model development, evaluation, and web application deployment. It includes Python scripts for data preprocessing, feature extraction, model training, and evaluation, along with the Streamlit application files used for the user interface. The repository also maintains the serialized Logistic Regression model, fitted TF-IDF vectorizer, and VADER scaler, which are required for classifying hotel reviews.

Git and GitHub are used for version control, enabling the tracking of code modifications, management of project updates, and maintenance of different development stages. The repository provides an organized structure for reviewing the implementation, identifying errors, and making future improvements. This version-controlled approach supports project maintainability, reproducibility, and further enhancement of the fake review detection system.

GitHub Repository

The project source code is maintained in a public GitHub repository, which contains the files required for model implementation and application deployment. The repository includes the trained model components, dataset, application code, and project documentation. GitHub is used to maintain version history and track modifications made during development.

Figure 7.1 shows the GitHub repository of the Fake Review Detection Using Machine Learning project, including its directory structure.

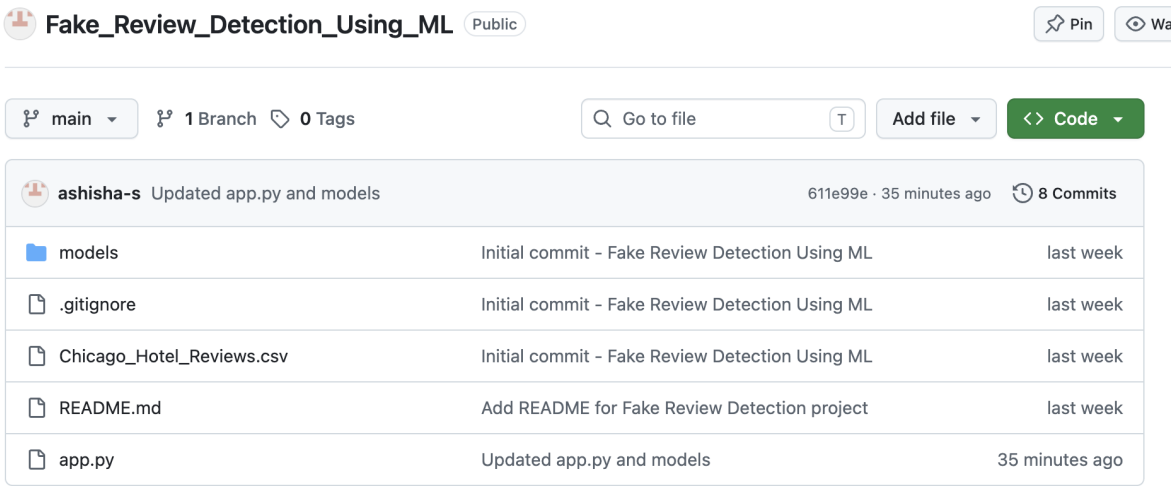


Figure 7.1: GitHub repository

8. Conclusions

In conclusion, the Fake Review Detection Using Machine Learning project successfully achieved its objective of developing a machine learning-based system for identifying deceptive and genuine hotel reviews. The system was developed using the Ott Deceptive Opinion Spam Corpus, containing 1,600 labelled hotel reviews with an equal distribution of fake and genuine reviews. Natural Language Processing techniques were applied for text preprocessing, while TF-IDF, linguistic features, and VADER sentiment analysis were used to extract relevant information from the review text.

Three supervised machine learning algorithms, namely Logistic Regression, Linear Support Vector Machine (SVM), and Random Forest, were implemented and evaluated using four different feature configurations. A total of twelve model–feature-set combinations were compared using Accuracy, Precision, Recall, F1-score, ROC–AUC, and Confusion Matrix. The experimental results demonstrated that TF-IDF provided a strong baseline for classification, while the addition of VADER sentiment features improved the performance of Logistic Regression and Linear SVM. However, the selected linguistic features did not provide further improvement under the current experimental setup.

Among the evaluated configurations, Logistic Regression with TF-IDF and VADER sentiment features achieved the highest Accuracy of 91.88% and F1-score of 91.93%. The selected model also achieved a Precision of 91.36%, Recall of 92.50%, and ROC–AUC of 96.81%. Based on its overall classification performance and ability to generate probability estimates, Logistic Regression with TF-IDF and VADER was selected for the final prediction system.

The trained model was integrated into a Streamlit-based web application to enable classification of user-submitted hotel reviews. The application processes the review text, extracts TF-IDF and VADER sentiment features, and generates a prediction indicating whether the review is fake or genuine, along with the corresponding classification probabilities. The interface also provides information about the extracted features and their contribution to the prediction, improving the transparency and usability of the system.

Overall, the project demonstrates the effectiveness of combining Natural Language Processing, sentiment analysis, and classical machine learning techniques for detecting deceptive online reviews. The experimental findings highlight the importance of selecting suitable feature combinations rather than simply increasing the number of features. Although the system was developed and evaluated using hotel reviews, it provides a foundation for future enhancements involving larger datasets, additional review domains, and more advanced classification techniques.

9. Future Work

- **Advanced Machine Learning Algorithms:** Future improvements can explore ensemble learning algorithms such as XGBoost and CatBoost, along with deep learning models such as LSTM and BERT, to improve classification accuracy and capture complex linguistic patterns.
- **Larger and Diverse Datasets:** The system can be trained on larger datasets containing reviews from e-commerce platforms, restaurants, and travel websites to improve generalization across different domains.
- **Advanced Feature Engineering:** Additional features such as semantic similarity, contextual embeddings, and readability scores can be incorporated to identify deceptive patterns more effectively.
- **Detection of AI-Generated Reviews:** The system can be extended to detect AI-generated deceptive reviews by incorporating appropriately labeled datasets containing machine-generated content.
- **Enhanced Explainability:** Explainable AI techniques such as SHAP and LIME can be integrated to provide more detailed interpretations of model predictions.
- **Multilingual Review Detection:** Future versions can support multiple languages through multilingual text preprocessing and language-specific feature extraction techniques.
- **Cloud Deployment and Platform Integration:** The Streamlit application can be deployed on cloud platforms and integrated with online review websites through APIs to enable scalable and automated review classification.

10. Appendix

10.1 Minimum Software Requirements

Operating System: Windows 10 / Linux / macOS

Python Version: Python 3.10 or above

Required Libraries:

- Streamlit (for web application development)
- NumPy (for numerical operations)
- Pandas (for data manipulation and analysis)
- Scikit-learn (for feature extraction, model training, evaluation, and classification)
- NLTK (for text preprocessing, lemmatization, stop-word removal, and VADER sentiment analysis)
- SciPy (for sparse feature matrix operations)
- Joblib (for saving and loading trained models, vectorizers, and scalers)
- Matplotlib / Seaborn (for exploratory data analysis and visualization)

Development Tools:

- Google Colab / Jupyter Notebook (for data preprocessing, feature extraction, model training, and evaluation)
- Visual Studio Code (for Streamlit application development)
- Git / GitHub (for version control and repository management)

Browser: Latest version of Google Chrome, Mozilla Firefox, or Microsoft Edge for accessing the Streamlit web application.

10.2 Minimum Hardware Requirements

- **Processor:** Dual-core 2.0 GHz processor or higher. A quad-core processor is recommended for faster data preprocessing and model training.

- **RAM:** Minimum 4 GB RAM, with 8 GB recommended for smooth execution of data analysis, machine learning operations, and the Streamlit web application.
- **Storage:** At least 2 GB of available disk space for the dataset, Python environment, trained model files, application code, and other project resources.
- **Display:** Standard display resolution of 1366×768 pixels or higher.
- **Internet Connection:** An internet connection is required for:
  - Installing Python libraries and dependencies.
  - Downloading the hotel review dataset.
  - Accessing Google Colab for model development and experimentation.
  - Accessing GitHub for version control and repository management.
  - Downloading required NLTK resources during initial setup.

11. References

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